SlyTherm Dual-Fuel Smart Thermostat

Installation & Commissioning Manual

Model ST-1 (Dettson dual-fuel edition)

SlyTherm

DRAFT v0.4 — PRE-RELEASE, NOT FOR FIELD USE

Revision	Date	Changes
0.3	2026-06-11	Pre-release draft: installation, wiring, CT-485 bus, relay stage, HA integration, commissioning, rollback
0.4	2026-07-06	SlyTherm branding; onboard isolated RS-485 (GPIO43/44); HA topic + roster validation

Contents

DRAFT v0.4 — PRE-RELEASE, NOT FOR FIELD USE

This manual describes a design that has not yet completed bench validation or field commissioning. Several behaviors of the installed equipment are still pending verification and are explicitly marked as such throughout. Do not install this product from this draft.

Product, model, and manufacturer names are provisional branding and may change before release.

About this manual

This manual is written for the **installing technician and systems integrator**. It covers physical installation, low-voltage wiring, the RS-485 bus connection, the relay output stage (where applicable), configuration, Home Assistant integration, commissioning, and rollback.

A separate **User Manual** covers day-to-day operation of the thermostat.

The SlyTherm is designed for one specific equipment family:

- **Furnace:** Dettson Chinook modulating gas furnace (ClimateTalk / CT-485 communicating control, 40–100 % modulation).
- **Heat pump:** Gree-built outdoor unit sold with the Dettson system — either a communicating Dettson Alizé configuration (with the Dettson K03085 interface board) or a conventional 24 V Gree FLEXX configuration.

Do not install the SlyTherm on any other equipment.

Who may install this product

Installation involves the low-voltage control system of a **gas-fired appliance** and a **heat-pump compressor circuit**. In many jurisdictions, work on gas-appliance controls is restricted to licensed technicians. Read Section 1 (Critical Safety Information) in full before opening any equipment. At minimum, a **licensed HVAC technician must review the completed installation** before the system is left in service.

Conventions used in this manual

Marker	Meaning
WARNING	Risk of personal injury, fire, carbon monoxide exposure, or significant property/equipment damage.
CAUTION	Risk of equipment damage or malfunction.
> ⚠ Pending verification:	A behavior of the installed equipment that the design depends on but that has **not yet been confirmed** on real hardware. Treat as unknown until the referenced commissioning step proves it.
> ⚠ TBD (Phase 0):	Content that depends on the result of the **pre-install equipment survey** (called “Phase 0” in engineering documents; see Section 2.4). The survey decides which installation case applies to your site.

Procedures are numbered and must be performed in order. Specifications and parameters are given in tables; temperatures are in °C unless noted.

Document set

Document	Audience
User Manual	Homeowner / daily operator
Technical Installation Manual (this document)	Installer / integrator

1. Critical Safety Information

WARNING — This product connects to the control system of a gas appliance and a heat-pump compressor. Read this section in full before energizing anything.

1.1 What this controller does — and does not — control

The SlyTherm **does not command the gas valve.** It sends a ClimateTalk (CT-485) **heat demand** — a capacity *request* — over the furnace's RS-485 bus. The furnace's certified **Integrated Furnace Control (IFC)** is the only device that opens or modulates the gas valve, and the IFC independently enforces every combustion safety regardless of anything received on the bus:

- Flame sensing and flame-failure lockout (no proven flame → gas valve closes, lockout).
- Primary high-limit (overtemperature → shutdown).
- Rollout switch and pressure/vent (draft) switch proving.
- Ignition-trial limits, retries, and hard lockout.
- Internal modulation limits — the furnace never fires above its rated input regardless of the demand it receives.

These interlocks are hard-wired or firmware-locked inside the certified IFC and are **not on the RS-485 bus**. They cannot be disabled over CT-485, and a missing or malformed demand message does **not** open the gas valve.

The same boundary applies on the heat-pump side: the controller requests capacity; the Gree-built outdoor unit's own controls execute it. Note, however, that the outdoor unit's protections (pressure trips, restart delay, fault latching) are **equipment protections, not a certified life-safety chain** — the SlyTherm's own compressor timers and lockouts are the primary protection layer (see Section 7).

1.2 Who may perform this work

- Work on gas-appliance controls is **restricted to licensed technicians in many jurisdictions**. Confirm local requirements before starting.
- Wiring at the furnace terminal strip, the cased-coil board, and the outdoor unit must be performed by, or under the supervision of, a **licensed HVAC technician**.

- The network/Home Assistant integration portion (Sections 8–9) may be performed by a competent systems integrator.
- A **licensed HVAC technician must review the completed installation** and witness the commissioning tests in Section 10 — in particular the per-channel loss-of-communications tests — before the system is left in service.

1.3 Certification, warranty, and insurance — stated honestly

- The furnace is certified (CSA, for Canada/US gas appliances) **as a system with an approved control**. The OEM communicating thermostat family (R02P032/R02P034; the original R02P030 is discontinued) is part of that certified configuration.
- Installing the SlyTherm in place of the OEM thermostat is a **modification of a certified gas appliance's control system**. Riding the bus *as a thermostat* — with the IFC keeping all combustion safeties — does not alter the certified combustion-safety path, but it **is outside the certified configuration**. The same logic applies on the heat-pump side: the SlyTherm stays on the thermostat side of the coil board / interface board and never touches the Gree-proprietary outdoor-unit link.
- **Honest implications:** this installation **can void the furnace and heat-pump warranties**, may **violate local gas or electrical code** (work on gas-appliance controls is often restricted to licensed technicians), and **could affect home insurance** after any fire, carbon-monoxide, or water incident traced to the controller.
- This is an **experimental control on a life-safety appliance, operated at the owner's risk**. If the owner is not willing to accept that, install a Dettson-supported thermostat (R02P034) and integrate at the Home Assistant level instead.

1.4 Mandatory conditions of installation

All of the following are required, not recommended:

1. **Do not remove or bypass any furnace or outdoor-unit safety wiring.** Tap the CT-485 bus / thermostat terminals only; leave the R/C/1/2 terminal semantics intact; never open the Gree H1/H2 or COND links.
2. **Install a carbon-monoxide (CO) alarm** in the dwelling and verify it works. This is a precondition of running the SlyTherm, not an option.
3. **Retain an OEM thermostat for rollback.** Keep a Dettson R02P034 (communicating) — or an R02P033 for a conventional installation — on site so the system can be returned to its certified configuration in minutes (Section 11). Design the wiring so that swap takes minutes.

4. **Have a licensed HVAC technician review the final installation** and confirm, per demand channel, that loss of communications drops the equipment call (Section 10).
5. **Document** that the installation does not alter the combustion-safety chain or the refrigerant-circuit protections.

1.5 Specific wiring hazards

> **WARNING — V/W2 terminal.** The Chinook's V/W2 terminal accepts a 24 V *modulating signal* only. **Never apply raw 24 VAC to V/W2** — this is a documented damage hazard. Take care during any probing or wiring not to short R to V/W2.

> **CAUTION — 24 VAC is AC, and furnace boards commonly half-wave-rectify with the C terminal shared.** Powering the SlyTherm from anything other than the specified **isolated** AC-input supply, or bonding the controller's DC ground to furnace C, can short a half-cycle of the transformer and blow the furnace's control fuse (Section 4).

> **WARNING — Reversing-valve polarity (relay installations).** The Gree convention is **B = energized in HEATING** — the opposite of the common "O = cool" default. A wrong polarity setting inverts heating and cooling. Polarity must be verified at commissioning and is only ever changed with the compressor idle (Sections 6 and 10).

1.6 Loss of heat is also a hazard

The SlyTherm's design fails toward **no demand** on any fault. The flip side is that a sustained fault in heating season means **no heat** — itself a hazard (frozen pipes, habitability). The controller raises an escalating alarm (on-screen and via Home Assistant) on sustained forced-no-demand; the recovery path is rollback to the OEM thermostat (Section 11). Brief the homeowner on both the alarm and the rollback procedure.

1.7 Stop conditions

Stop the installation and do not proceed if:

- Equipment bus captures ever show the thermostat sending a **raw valve duty** that the IFC obeys without independent flame/limit supervision. That would indicate an uncertified, unsafe control path and is out of scope for this product.

- Any commissioning test in Section 10 shows that a demand channel does **not** drop to zero on loss of communications and no hardware means of removing that call exists.
- The pre-install survey (Section 2.4) finds equipment outside the supported family.

2. System Overview and Pre-Install Survey

2.1 What you are installing

The SlyTherm is a **single wall-mounted touchscreen controller** installed at the location of the OEM thermostat. The existing thermostat wall plate already carries the four conductors the controller needs in a communicating system:

Wall terminal	Function
R	24 VAC hot (from the furnace transformer)
C	24 VAC common
1	CT-485 bus, Data A (+)
2	CT-485 bus, Data B (−)

Behind the display, inside the wall enclosure, live the supporting electronics — all required, none optional:

- **Isolated AC-input power supply** (24 VAC → 5 V) with inline fuse and MOV (Section 4).
- **3.3 V RS-485 transceiver** with explicit driver-enable (DE/RE) control and transient protection (Section 5).
- **External hardware watchdog** that forces the controller to a no-demand state if the processor hangs (Section 7).
- Local fallback temperature sensor (DS18B20, 1-Wire).

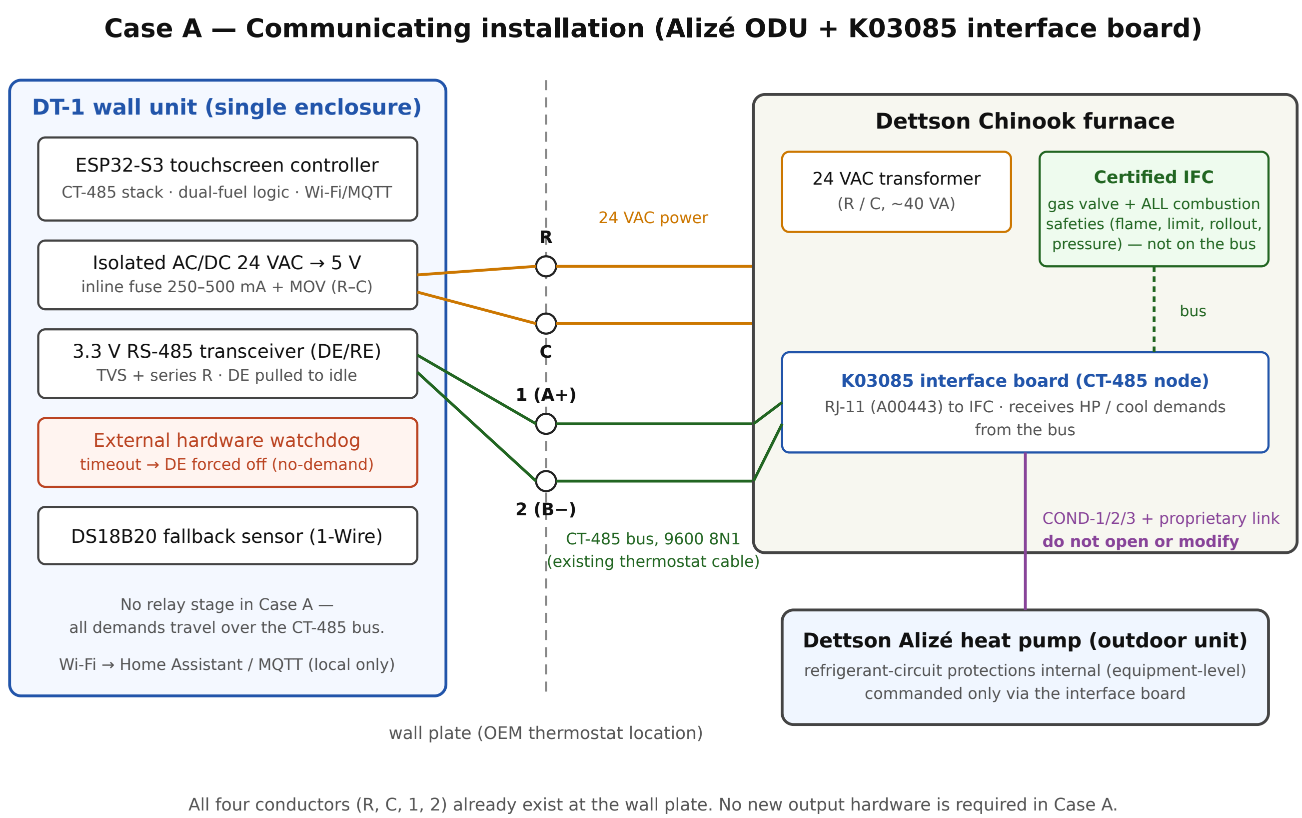
The controller speaks **ClimateTalk (CT-485)** to the furnace for modulating gas demand (40–100 %), supervises the dual-fuel arbitration between gas and heat pump, and integrates with **Home Assistant** over MQTT (local network only — no cloud service).

2.2 The two installation cases

How the heat pump is commanded depends on which control architecture is installed at the site. **The installation forks here**, and the pre-install survey (Section 2.4) is how you determine which case applies. Do not assume either case, and do not purchase Case-B hardware before the survey.

Case A — Communicating (Dettson Alizé outdoor unit + K03085 interface board)

The outdoor unit is reached *via the bus*: a Dettson K03085 interface board inside the furnace (RJ-11 cable to the IFC; COND-1/2/3 plus a Gree-proprietary link to the outdoor unit) is a CT-485 node, and the SlyTherm sends heat-pump and cooling demands to it over the same two bus wires already at the wall. **No additional output hardware is required.** The 24 V sense inputs described in Section 6 are still recommended as permanent instrumentation.

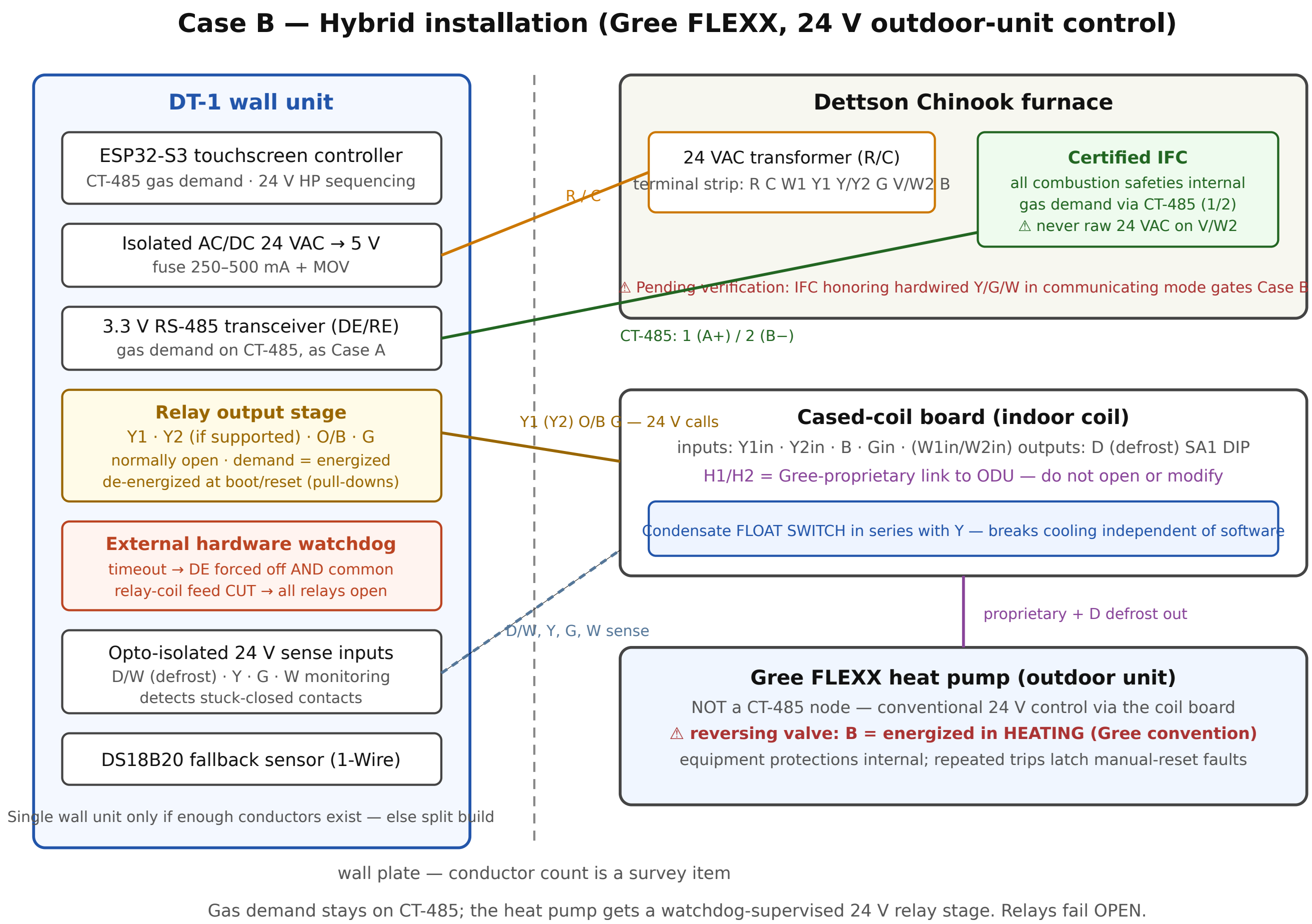


Case A — communicating installation, block diagram

Case B — Hybrid (true Gree FLEXX, conventional 24 V outdoor-unit control)

A true Gree FLEXX outdoor unit is **not a CT-485 node** — it is commanded by conventional 24 V signals at the cased-coil board (its H1/H2 RS-485 link is Gree-proprietary and must never be touched). In this case, gas demand stays on CT-485, and the SlyTherm adds a **24**

V relay output stage (Y1, Y2 if supported, O/B, G) plus opto-isolated sense inputs and a hardwired condensate float switch (Section 6).



Case B — hybrid installation, block diagram

> ⚠ **Pending verification (gates Case B):** it has not yet been confirmed whether the Chinook IFC honors hardwired Y/G/W signals **while operating in communicating mode**, or whether bus communications disable the 24 V terminals (the furnace's cooling-CFM DIP switches are documented as ignored in communicating mode). The hybrid path is viable only if this mixed mode works. This must be proven on the installed equipment before Case B wiring is committed.

> ⚠ **Pending verification (Case B alternative gas path):** the Chinook's V/W2 terminal accepts a 24 V modulating signal providing the full 40–100 % range with no CT-485 connection at all. It exists as a documented fallback path only. **Never apply raw 24 VAC to V/W2** — documented damage hazard.

Single unit vs. split installation

The standard installation is a single wall unit. Two findings force a **split installation** instead (a headless controller at the furnace/coil, with the wall display — or a Home Assistant dashboard — connected over the network only):

1. **Case B with too few conductors at the wall plate.** A wall unit can only drive Y1/Y2/O-B/G if enough conductors exist in the existing thermostat cable. Count them during the survey.
2. **Bus-timing qualification failure.** Single-unit operation is gated on a bench measurement showing the touchscreen firmware meets the CT-485 bus-timing budget. This is a recorded pass/fail from bench testing, not a field judgment.

> ⚠ **TBD (Phase 0):** the installed architecture (Case A vs Case B) and the single-vs-split decision are outputs of the pre-install survey and bench qualification. This draft cannot tell you which case you have.

2.3 What travels where (summary)

Signal	Case A	Case B
Gas heat demand (40–100 % modulating)	CT-485 bus	CT-485 bus (V/W2 modulating signal exists as fallback — pending verification)
Heat-pump heat / cool demand	CT-485 bus (to interface-board node)	24 V relays: Y1, Y2 (if supported), O/B, G
Blower/fan demand	CT-485 bus (mapping pending verification)	CT-485 bus and/or G relay (mapping pending verification)
Defrost indication	bus signature (pending verification)	D/W 24 V sense input
Furnace status, modulation %, faults	CT-485 bus	CT-485 bus
Condensate protection	hardwired float switch (required whenever cooling is enabled, either case)	hardwired float switch in series with Y

2.4 Pre-install survey (required before any purchase or wiring)

The survey is a **zero-risk paper-and-camera exercise** — no electronics are connected, nothing is energized beyond normal equipment operation. Its result decides Case A vs Case B and single vs split. (Engineering documents call this survey “Phase 0”.)

> **WARNING:** during the survey, never apply 24 VAC to the V/W2 terminal, and do not disconnect any safety wiring. Photograph; do not rewire.

Procedure:

1. **Record the existing wall thermostat model.**
    - R02P032 or R02P034 → communicating installation (expect Case A).
    - R02P033 or any conventional thermostat → conventional installation (expect Case B).
2. **Open the Chinook blower door and look for an interface board.** A K03085 (or K03081) interface board is identifiable by its RJ-11 cable (A00443) to the IFC and COND-1/2/3 wires running toward the outdoor unit. Present → Case A. Absent → Case B.
3. **Read the outdoor-unit nameplate.** Record the model (MHD/Alizé = communicating family; GUD/FXD FLEXX = conventional family) and the rated minimum operating ambient (needed for configuration, Section 8).
4. **Record the IFC DIP switches S4-2 / S4-3.**
    - OFF/OFF = modulating/communicating mode.
    - ON/ON = 2-stage thermostat mode.
5. **Photograph every low-voltage conductor** at:
    - the furnace terminal strip (R, C, W1, Y1, Y/Y2, G, V/W2, B);
    - the cased-coil board (Y1in/Y2in/W1in/W2in/Gin, B, D, H1/H2, and the SA1 DIP switch);
    - the outdoor unit’s low-voltage terminals.
6. **Count the conductors in the thermostat cable at the wall plate.** Note total conductors, spares, and condition. Case B as a single wall unit needs R, C, 1, 2 **plus** Y1/(Y2)/O-B/G and any sense lines — if the cable cannot carry them, plan the split installation.
7. **Document the result:** architecture (A or B), thermostat model, outdoor unit model and rated minimum ambient, IFC DIP states, the full low-voltage wiring map, and the

wall-plate conductor count.

Survey decision table:

Finding	Conclusion
Communicating thermostat + interface board + Alizé outdoor unit	**Case A** — no relay stage; all demands over the bus
Conventional thermostat, no interface board, FLEXX outdoor unit	**Case B** — relay stage required; mixed-mode verification required first
Mixed/ambiguous findings	Stop. Resolve with Dettson documentation before proceeding
Case B and insufficient wall conductors	**Split installation** (relay stage at the furnace/coil end)

3. Mounting

3.1 Location

Mount the SlyTherm wall unit at the **existing OEM thermostat location**. This is deliberate: the wall plate there already terminates R/C/1/2 (24 VAC power and the CT-485 bus), and the location was chosen by the original installer for representative room air. Do not relocate the controller to a wall cavity with poor air circulation, direct sunlight, or proximity to supply registers — the on-board/local temperature sensing is a safety fallback and must read plausible room air.

Standard thermostat-placement practice applies: interior wall, approximately 1.5 m above the floor, away from drafts, sun, lamps, and appliances.

3.2 Wall bracket and enclosure

1. Remove the OEM thermostat from its base. **Keep the thermostat and its base intact** — it is your certified rollback device (Section 11).
2. Label every conductor at the wall plate before disconnecting anything (R, C, 1, 2, and any extras found in the survey).
3. Install a **low-voltage old-work bracket** in the existing opening. The bracket carries the SlyTherm wall enclosure; no electrical box is required for Class-2 wiring, subject to local code.
4. The SlyTherm wall enclosure must house, behind the display:
    - the isolated AC/DC converter module (Section 4);
    - the inline fuse and MOV;
    - the RS-485 transceiver carrier board with TVS protection;
    - the external hardware watchdog;
    - terminal blocks for the field wiring.
5. Use **ferrules on all stranded conductors** entering screw terminals, and strain-relieve all wiring. Thermal cycling loosens bare strands over time.
6. **Fully enclose the AC/DC converter module** — no exposed mains-style terminals or bare boards in the wall cavity.

> ⚠ **Pending verification (local code):** Class-2 24 VAC wiring in a wall cavity is generally permissible, but the code treatment of an **AC/DC converter module mounted in-wall** must be confirmed for your jurisdiction before the enclosure design is finalized.

3.3 Enclosure material and temperature

Location	Requirement
Wall enclosure (living space)	Flame-retardant enclosure (UL94 V-0 rated plastic or polycarbonate preferred); screw terminals; strain relief
Furnace-side enclosure (split installation or Case-B relay stage at the coil)	**Polycarbonate or equivalent rated ≥ 105 °C near heat.** ABS is only ~80 °C continuous and furnace cabinets exceed that near the burner — mount on the **cool side of the furnace, outside the cabinet**
Electrolytic capacitors (any enclosure)	105 °C-rated parts

3.4 Furnace-side junction point

In a single-unit Case A installation there is no controller at the furnace; an enclosure at the furnace is optional and serves only as a junction or instrumentation point. In a **split installation** or a **Case B installation with the relay stage at the coil**, the furnace-side enclosure houses the headless controller and/or the relay and sense modules — apply the temperature rules above.

3.5 Sensor placement

- **Local fallback sensor (DS18B20):** in a single-unit wall installation the sensor is at the wall — in living space, which is ideal. In a split installation, do **not** place it adjacent to the furnace plenum if avoidable; the 1-Wire bus tolerates long runs, so relocate it into living space. A plenum-adjacent fallback sensor is a floor-keeper, not a comfort input, and forces a degraded operating mode if it ever becomes the only source.
- **Outdoor temperature sensor (wired DS18B20):** mount on a **north wall, shaded**, with an outdoor-rated cable run. This sensor feeds the dual-fuel balance point and compressor lockouts (Section 8); a sensor in sun or over a dryer vent will corrupt changeover decisions.

- **Optional supply-air sensor (DS18B20):** in the supply trunk if the coil-freeze guard option is used (cooling dropped if supply air falls below approximately 5 °C).

4. Power

4.1 Source and budget

The SlyTherm is powered from the furnace’s 24 VAC control transformer via the **R** and **C** conductors at the wall plate.

Item	Value
Nominal supply	24 VAC RMS (≈ 34 V peak)
Real-world supply	HVAC transformers run hot: 28–30 VAC at light load → **40 V+ peak**. The converter front end must be rated ≥ 50 V input
Transformer capacity	Typically 40 VA with little headroom
SlyTherm draw	Display-class board: 1.6–2.25 W verified; **≤ ~3 VA** from the transformer including conversion losses
Transients	Processor + Wi-Fi peaks ~500 mA @ 3.3 V, 600 mA+ transient; display backlight inrush at power-up

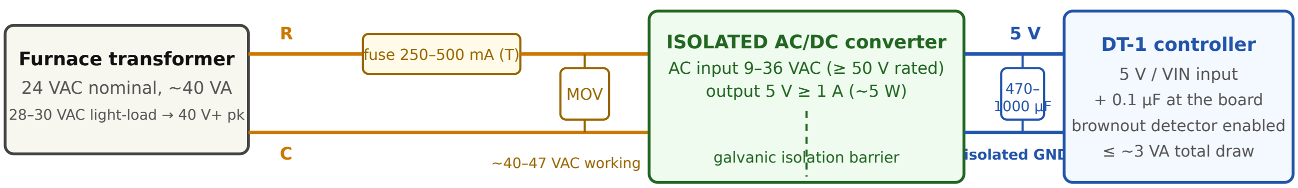
4.2 Isolated supply — mandatory

CAUTION: 24 VAC is **AC** — a DC-DC buck converter cannot take it. Worse, HVAC control boards frequently **half-wave-rectify with C shared**: full-wave-bridging the same transformer (or bonding your DC ground to C) can short a half-cycle and blow the furnace’s control fuse.

Use an **isolated AC-input AC/DC converter module**: input rated 9–36 VAC (≤ 50 VDC), output 5 V at ≥ 1 A (~5 W). Isolation eliminates the shared-C short outright.

CAUTION: many inexpensive AC-input modules are transformerless and **non-isolated**. The production wall installation requires a **confirmed-isolated** supply. If no suitable isolated AC/DC module is available, the fallback is: bridge rectifier (≥ 100 V) → bulk capacitor (≥ 470 µF / 63 V) → isolated DC-DC converter — and even then, **never bond the DC ground to furnace C anywhere**.

Power chain — 24 VAC wall supply to the controller



Rules that protect the furnace:

- Isolation is mandatory — furnace boards often half-wave-rectify with C shared; a non-isolated converter or a DC-GND-to-C bond can short a half-cycle and blow the furnace control fuse.
- Do NOT bond the controller's DC ground to C anywhere. The only bus-side reference is via the RS-485 barrier (or ~100 Ω) — see the bus section.
- Fuse stays well below the furnace's own control fuse: a fault blows this fuse, never the furnace's. Backlight inrush may need 500 mA instead of 250 mA.

Power chain — fuse, MOV, isolated converter, controller

4.3 Input protection — mandatory

Component	Specification	Purpose
Inline fuse on **R**	Slow-blow **250–500 mA**, 5×20 mm holder	Well below the furnace's own control fuse, so a fault blows *this* fuse, never the furnace's. The display backlight inrush may nuisance-trip a 250 mA fuse — be prepared to fit 500 mA (still far below the furnace control fuse)
MOV across **R–C**	~40–47 VAC working voltage	Clamps gas-valve / inducer / igniter switching transients
Bulk capacitance at the controller	470–1000 µF (63 V, 105 °C) + 0.1 µF ceramic, at the 5 V input	Kills Wi-Fi-transient brownouts

The controller's internal brownout detector remains **enabled**; on brownout the firmware boots to a no-demand state (Section 7).

4.4 Power wiring table

From	To	Notes
Wall plate **R** (24 VAC hot)	Isolated AC/DC **IN+**	via inline fuse, 250–500 mA slow-blow
Wall plate **C** (24 VAC common)	Isolated AC/DC **IN−**	MOV connected across R–C
AC/DC **5 V OUT+**	Controller **5 V / VIN**	bulk capacitors at the controller
AC/DC **5 V OUT−**	Controller **GND**	**isolated DC ground — do NOT bond to C**

The only deliberate reference between the controller's electronics and the furnace's 24 V system is the RS-485 bus-side ground reference described in Section 5.4 — nothing else may bridge the isolation barrier.

4.5 Power-up behavior

On power-up (or any reset) the controller:

- holds the RS-485 driver disabled (the bus stays silent through boot — verified by oscilloscope at commissioning, Section 10);
- holds all relay outputs de-energized (Case B);
- asserts **no demand on any channel** until mode, setpoints, and sensors are validated;
- enforces the compressor post-boot hold-off if compressor-timer state was lost (Section 8).

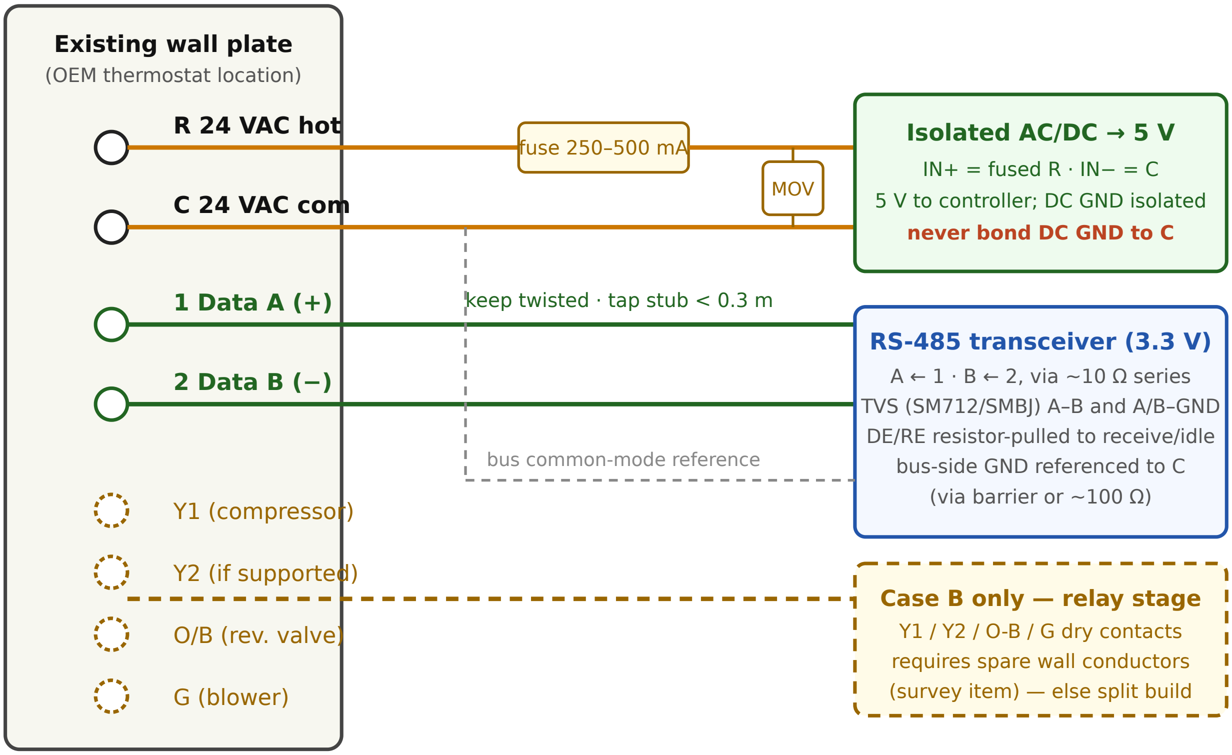
5. CT-485 Bus Connection

5.1 Bus basics

Parameter	Value
Physical layer	EIA/TIA-485, 2-wire half-duplex (“CT-485”)
Wall terminals	**1 = Data A (+)**, **2 = Data B (−)**
Signaling	9600 bps, 8N1 (confirm by capture at commissioning)
Frame delimiting	inter-byte idle gap > 3.5 ms (no preamble byte)
Topology	coordinator-polled (token); the thermostat transmits only in its slot

Keep the bus pair **twisted** all the way to the transceiver, and keep the tap stub **shorter than ~0.3 m**.

Wall-plate field wiring



No added 120 Ω termination or bias until measured AT THE WALL PLATE — the OEM location may be a terminated bus end.
Ferrules on all stranded conductors. Label every wire before disconnecting the OEM thermostat.

Wall-plate wiring — R/C power and CT-485 bus to the SlyTherm wall unit

5.2 Transceiver requirements

The SlyTherm board’s transceiver (on the carrier board behind the display) must meet all of these — substitutions that miss any one of them break the safety chain:

- **3.3 V-native** part (THVD1410/THVD1450 or MAX3485 class). **Never a MAX485** — it is a 5 V part whose receive output over-volts the controller’s 3.3 V inputs.
- **Explicit DE/RE (driver-enable) control** — no auto-direction modules. The external hardware watchdog must be able to force the driver off (Section 7), and an auto-direction module gives it no pin to do that, nor a guaranteed-silent idle.
- **DE/RE pulled to the receive/idle state by a resistor**, so a booting, resetting, or crashed controller **releases the bus and stays silent**.
- Powered from the controller’s clean 3.3 V rail.

> **Onboard transceiver.** The SlyTherm carrier board satisfies these requirements on-board: it provides a **galvanically-isolated RS-485 transceiver** whose A/B data lines are driven from the ESP32-S3 UART0 on **GPIO43 (TX)** and **GPIO44 (RX)**, with the DE/RE direction control on a controller GPIO that the external hardware watchdog can force off (Section 7). Because the transceiver is isolated, the isolation requirement of Sections 4 and 5.4 is met at the board — no external isolator is needed.

5.3 Line protection

Component	Placement
Bus-rated TVS array (SM712, or SMBJ pair)	A–B, and A/B to bus-side ground
~10 Ω series resistors	in the A and B lines into the transceiver

5.4 Bus ground reference

RS-485 requires a common-mode reference (−7 to +12 V window). With the isolated power supply of Section 4, give the bus side a defined reference: tie the transceiver's **bus-side ground to furnace C** — directly on the bus side of the isolation barrier, or through ~100 Ω if the design is non-isolated at that point.

> **CAUTION:** do **not** hard-bond the controller's DC ground to C — that recreates the half-wave short described in Section 4.2.

5.5 Termination and bias — measure first, at the wall plate

You are adding a node to an **already-terminated, already-biased** 2–3-node bus. Blindly adding a 120 Ω terminator in the middle of the bus overloads the drivers and can break communications for every node. Likewise, the furnace almost certainly biases the idle bus — adding more bias can skew it.

The OEM thermostat location **may have been a terminated bus end** — if your node physically becomes the bus end at the wall, the answer differs from a mid-bus tap. That is why this measurement is performed **at the wall plate**, not assumed from the furnace end.

Procedure (at the wall plate, before connecting the SlyTherm's A/B lines):

1. With the system powered **off**, measure the DC resistance between terminals 1 and 2 (A–B).
    - ≈ 120 Ω suggests one terminator is reachable from here (you may be at an unterminated end of a singly-terminated bus);
    - ≈ 60 Ω suggests both terminators are present elsewhere — the bus is fully terminated and you are a mid-bus tap. **Add nothing.**
2. With the system powered **on** and the bus idle, measure the differential idle voltage |V(A−B)|.
    - ≥ ~200–300 mV: the bus is adequately biased. **Add no bias.**
    - < ~200–300 mV: note it; bias may be required at this node.
3. **Add a 120 Ω terminator only if** the measurements show your node has physically become a bus *end* that lacks termination.
4. **Add pull-up/pull-down bias only if** step 2 showed an inadequately biased idle bus.
5. Record all measurements and what (if anything) was fitted — they are part of the commissioning record (Section 10).

5.6 Connection table

From	To	Notes
Wall plate **1** (Data A+)	Transceiver **A**	TVS + ~10 Ω series; twisted pair; stub < 0.3 m; no added 120 Ω/bias until measured (5.5)
Wall plate **2** (Data B−)	Transceiver **B**	as above
Furnace **C** (via wall plate C)	Transceiver bus-side **GND**	common-mode reference (5.4)
Transceiver **RO**	Controller UART RX	3.3 V native — no divider
Transceiver **DI**	Controller UART TX	
Transceiver **DE+RE**	Controller GPIO **and** watchdog cut path	resistor pull to receive/idle; watchdog can force off (Section 7)

5.7 Boot silence and bus discipline

The CT-485 bus carries the furnace's own control traffic; corrupting it is a failure mode in its own right. The SlyTherm's bus discipline (all enforced in firmware, verified at commissioning):

- **Silent at boot/reset** — DE is resistor-pulled off; the oscilloscope check in Section 10 proves no byte is emitted during a reset.
- **Transmit only when granted the bus** by the coordinator's polling (token) mechanism; release the driver at idle.
- Driver pre/post-enable hold of ~300 µs around each transmission; 100 ms inter-packet idle respected.
- Every frame checksummed; persistent inability to obtain clean communications → the controller goes passive (no demand).

> ⚠ **Pending verification:** the exact demand-message payload offsets and the coordinator token timing budget have **not yet been confirmed from captures of the installed equipment**. Active transmission is gated on that confirmation; an installation from this draft must not enable the transmit path.

> **WARNING — one thermostat only.** The OEM communicating thermostat must be **removed from the bus** before the SlyTherm ever transmits. Two masters on a coordinator-polled bus is undefined behavior. (Keep the OEM thermostat on site for rollback — Section 11.)

6. Case B Wiring — 24 V Relay Outputs and Sense Inputs

This section applies **only to Case B** (hybrid installation: gas demand on CT-485, heat pump on conventional 24 V signals). Skip it entirely for Case A — except 6.5 (condensate float switch), which is required **whenever cooling is enabled, in either case**, and 6.4 (sense inputs), which is recommended instrumentation in both cases.

> ⚠ **TBD (Phase 0):** do not order or install Case-B hardware until the pre-install survey confirms the conventional/hybrid architecture.

> ⚠ **Pending verification (gates all of Case B):** whether the Chinook IFC honors hardwired Y/G/W while in communicating mode is unconfirmed (see Section 2.2). Prove mixed-mode operation on the installed equipment before committing this wiring.

6.1 Relay output stage

Output	Function	Contact requirement
Y1	Compressor stage 1	24 VAC-rated dry contact (relay or optotriac), normally open
Y2	Compressor stage 2 — only if the installed model supports two stages	as above
O/B	Reversing valve	as above
G	Indoor blower	as above

Rules (all mandatory):

1. **Normally open, demand = energized.** De-energized = no demand, always.
2. **Relays must idle de-energized at boot and reset.** Drive GPIOs carry external pull-downs so a booting or crashed controller cannot close a contact — the same discipline as the bus driver-enable line. Verified by scope/sense input at commissioning.

3. **The common relay-coil feed is switched by the external hardware watchdog, not by a GPIO.** On watchdog timeout, power to *all* relay coils is cut (Section 7). Rationale: "bus silence = safe" does **not** transfer to relays — a hung controller with Y latched closed would run the compressor indefinitely, with no refresh timer on a closed contact.

> **WARNING — Reversing-valve polarity.** The Gree convention is **B = energized in HEATING** — the opposite of the common "O = cool" default. Getting this wrong inverts heating and cooling. The polarity is verified on the installed equipment at commissioning (Section 10), and the O/B output changes state **only with the compressor proven idle** — never under load.

Firmware relay sequencing. The firmware drives these four outputs through a dedicated sequencer whose rules the commissioning interlock tests (Section 10) verify at the terminals:

1. **O/B changes only with the compressor proven idle** — its own Y output off *and* the compressor minimum-off timer (Section 8.1) served. A demand that needs the opposite valve position drops Y first and waits; when idle, the valve is pre-positioned from the system mode so it flips unloaded ahead of the next call, not in front of one.
2. **G is forced on with any Y** — the blower runs whenever the compressor is called, regardless of the fan demand. (Blower-*proven* remains a sense-side interlock on the G feedback input, 6.4: no Y without proof.)
3. **Y2 is only ever energized together with Y1.**
4. **Successive output transitions are spaced at least 500 ms apart** (`kRelayMinTransitionMs`, Section 8) so contacts never chatter; a deferred change waits at most one spacing window.
5. **All outputs are off at boot and on any failsafe drop.** The failsafe all-off is immediate — never deferred by the transition spacing — and the watchdog coil-cut path (Section 7) backstops it.

The defrost D-wire sense input is observation only: the outdoor unit owns its defrost cycle, and these outputs are not altered by it (defrost *tempering* reaches the furnace over the bus, never through these relays).

6.2 Conductor allocation at the wall plate

A single-wall-unit Case B installation must carry, in the existing thermostat cable: **R, C, 1, 2** (power + bus) **plus Y1, (Y2), O/B, G** and any sense conductors. If the survey's conductor count is insufficient, install the relay/sense stage at the furnace/coil end instead (split installation) — do not run new cable through uninspected cavities without local-code review.

6.3 Where the 24 V signals land

Wire the relay outputs to the corresponding inputs at the **cased-coil board / furnace terminal strip** exactly as a conventional thermostat would (Y1in/Y2in, B, Gin per the survey's photographed wiring map). Do not open or modify the Gree H1/H2 link or the COND wiring.

6.4 Opto-isolated 24 V sense inputs

Permanent instrumentation, recommended in both cases and required in Case B:

Sense input	Source	Purpose
D/W (defrost)	Outdoor unit's defrost output / furnace W node	Detects an active defrost so the controller can hold steady through it. **Note:** the thermostat's W2/AUX conductor ties to the same furnace-W node — the sense input must tolerate *either* source energizing the line
Y	Compressor call terminal	Output-vs-command mismatch detection (stuck relay)
G	Blower call terminal	Blower-proven interlock feedback
W	Gas/aux heat terminal	Instrumentation, mismatch detection

A sensed mismatch between commanded and actual terminal state raises an alarm and engages the watchdog coil-cut path (Section 7).

6.5 Condensate float switch — hardwired, both cases

> **Required whenever cooling is enabled, in either installation case.**

The float (wet-switch) in the condensate pan/drain must **break the cooling call independently of software**:

- **Case B:** wire the float switch **in series with the Y circuit** — a tripped float physically opens the compressor call no matter what the controller does.
- **Case A:** there is no thermostat Y conductor; the float switch must break the cooling call in the equipment's own cooling-call circuit instead.
- An additional parallel connection to a controller sense input is for **alarming only** — it never replaces the hardwired series break.
- Verified at commissioning: the float switch must kill the cooling call with the firmware unaware (Section 10).

> ⚠ **Pending verification (Case A break point):** the exact hardware break point for the float switch in a communicating installation (e.g. at the cased-coil board's cooling-call input) is finalized during installation engineering, against the photographed wiring map from the survey. Whatever point is chosen, it must pass commissioning test **E2** (Section 10): the float must kill the cooling call with the firmware unaware.

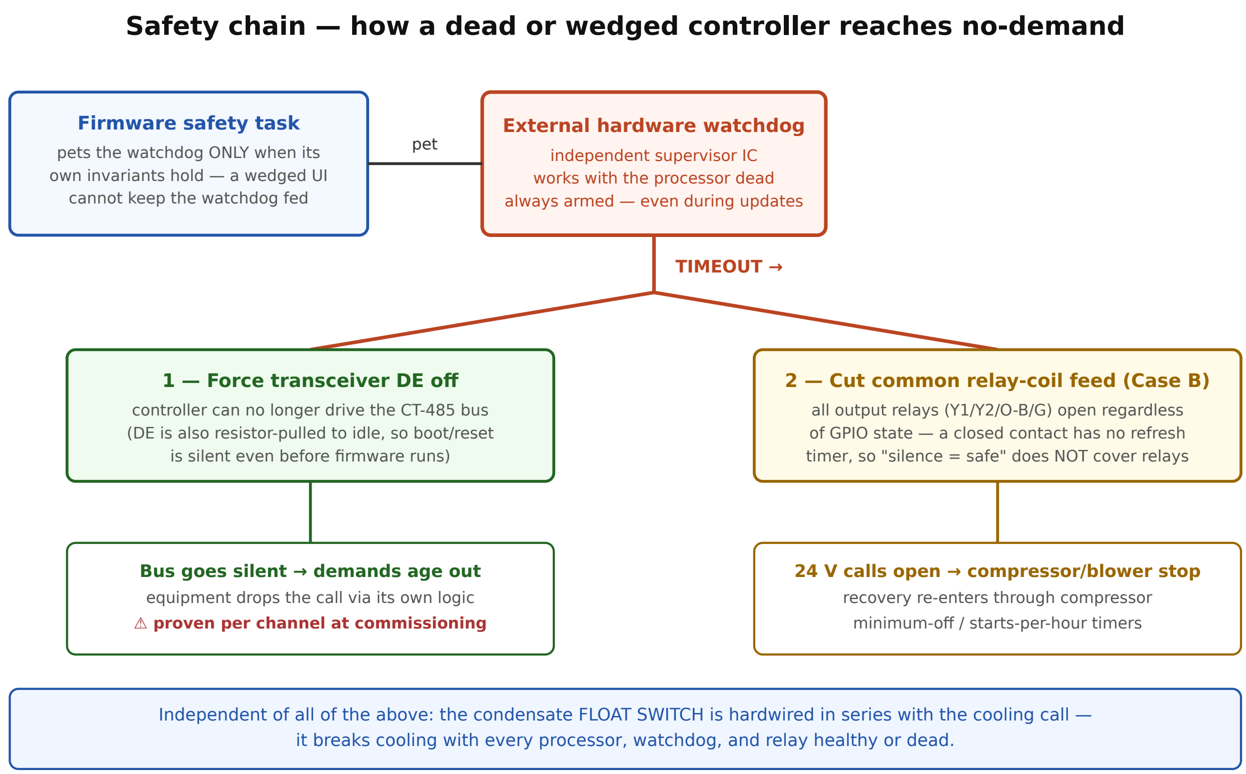
6.6 V/W2 modulating-signal path (fallback only)

The Chinook's V/W2 terminal accepts a 24 V **modulating signal** giving the full 40–100 % gas range with no CT-485 connection (the R02P029/1F95M-class thermostat path). It is documented here as a contingency only and is not the standard installation.

> **WARNING:** **Never apply raw 24 VAC to V/W2.** This is a documented damage hazard. If this path is ever used, only the proper modulating signal source may drive the terminal.

7. The Safety Chain

This section explains, for the installer, how the system fails safe — what the certified equipment enforces on its own, what the SlyTherm enforces, and what the external hardware enforces even when the SlyTherm's processor is dead. Understanding this is a prerequisite for the commissioning tests in Section 10.



Safety chain — watchdog forcing no-demand via driver-enable and relay-coil cuts

7.1 Division of enforcement

Layer	Enforces	Independent of the SlyTherm?
Furnace IFC (certified)	Flame sensing/lockout, primary high-limit, rollout, pressure/vent proving, ignition-trial limits, modulation ceiling	**Yes** — not on the bus, cannot be disabled over CT-485
Heat-pump outdoor unit (equipment protections, not a certified chain)	High/low refrigerant pressure trips, high discharge temperature, compressor overload, inverter-module protection, internal ~3-minute restart delay, minimum run time. Repeated trips **latch a manual-reset fault**	Yes — but treat as a **backstop only**. Note: smaller units have *no* low-pressure switch, and these are equipment protections, not life-safety. Verify the installed model's protections against its manual during the survey
External hardware (watchdog, pull resistors, float switch)	No-demand on processor hang; bus silence at boot; relays open at boot; cooling call broken on condensate overflow	Yes — works with the processor dead
SlyTherm firmware	Everything in 7.3	No — which is why the layers above exist

Relying on the outdoor unit's 3-minute delay as the anti-short-cycle mechanism is **not acceptable**: repeated trips latch manual-reset faults that strand the system. The SlyTherm's compressor timers are the primary protection.

7.2 Fail-to-no-demand, explained for installers

The SlyTherm's prime directive: **on any abnormal condition, fail to NO-DEMAND on all channels** — heat, cool, fan, backup/aux, defrost — and, in Case B, all relays de-energized — **without violating compressor minimum timers on the way down or back up**.

Why "no demand" is the safe direction:

- On the bus, equipment treats **loss of the thermostat's periodic demand messages as "drop the call"** — a silent node ramps the equipment down via its own normal logic. The failsafe is "go quiet," not "send an off command and hope it arrives."
- On relays, the equivalent is an **open contact**. But relays have the opposite failure polarity — a *stuck-closed* contact keeps equipment running — which is exactly why the watchdog cuts relay-coil **power** rather than asking firmware to open the relay.

> ⚠ **Pending verification (commissioning closes this):** "silence drops the call" is inferred from prior art for furnace heat and must be **proven per demand channel** on the installed system — including whether silence propagates to the non-coordinator unit (interface board / outdoor-unit side) in Case A. The pull-bus tests in Section 10 exist for this. Any channel where silence does **not** drop the call needs a hardware call-removal path before the system may be left in service — and that finding materially raises project risk (Section 1.7).

A flapping fault must not short-cycle the compressor through the failsafe path itself: recovery from any failsafe trip passes through the compressor minimum-off timer like any other start.

7.3 What the SlyTherm firmware enforces

- **Compressor protection (primary):** minimum OFF time, minimum ON time, maximum starts/hour, post-boot hold-off — with timer state persisted so a reboot cannot erase a pending minimum-off. If persisted state is missing at boot, the full hold-off is enforced.
- **Mutual exclusion of gas heat and compressor heat** — prohibited by the manufacturer (the heat-pump coil sits downstream of the heat exchanger; gas-heated air entering the coil drives head pressure toward the high-pressure cutout). Nothing at the bus level enforces this; the SlyTherm does, at its single demand-emission point. Defrost tempering is the sole sanctioned overlap, with a fixed demand and a 15-minute hard cap.
- **Low-ambient compressor lockout and dual-fuel balance-point logic**, with a hard configuration rule: any lockout/balance-point combination that leaves some outdoor-temperature band with **no permitted heat source** is rejected as invalid.
- **Sensor fault handling:** range, staleness, stuck-value, and divergence checks on every temperature input; loss of all valid indoor sources → demand zero on all channels + alarm. Unknown outdoor temperature → fail cold: gas allowed, compressor locked out.

- **Blower-proven interlock:** no cooling call without blower confirmation (bus telemetry in Case A; sensed G/Y feedback in Case B); the cooling call is dropped if feedback disappears.
- **Boot validation:** every boot starts at no-demand until mode, setpoints, and sensors validate; the restored mode is cross-checked against outdoor lockouts before any demand.
- **Reset-loop lockout:** ≥ 3 watchdog/brownout resets within 30 minutes → latched NO-DEMAND requiring manual clearing — a reboot loop must not repeatedly restart the compressor.
- **Runtime policy:** gas call > 4 h without progress → drop + alarm. Heat pump: **progress alarm only, never auto-cycled** — continuous near-balance-point running is normal for an inverter heat pump.
- **Bus discipline:** transmit only when polled, checksum everything, go passive if clean communications cannot be maintained.

7.4 The firmware safety supervisor

The enforcement points in 7.3 are coordinated by a dedicated **safety supervisor** in the firmware: every control cycle it is fed a set of health facts (sensor validity, control-loop liveness, demand-state sanity, bus and MQTT liveness, setpoint freshness) and decides three things — whether the external watchdog gets petted, whether demands are permitted at all, and which alarms are raised.

Watchdog petting is invariant-gated, and the invariants are not all equal:

- **Mandatory invariants** — a valid indoor temperature, a ticking control loop, and a sane demand state (emitted demands match commanded, no conflicting calls). If **any** of these is false, the supervisor **stops petting** the external watchdog, which then forces no-demand in hardware (Section 7.5).
- **Advisory invariants** — MQTT/Home Assistant connectivity and setpoint freshness. Loss raises an **alarm only**; petting continues. This is deliberate and binding: **loss of HA or the broker must never cause a no-heat event** (stale setpoints fall back to the bounded local profile, Section 8.9) — HA is a comfort layer, never a safety layer.
- **Bus silence (comms-loss deadman)** is the special case: if the CT-485 bus goes continuously silent past the deadman window (default 30 s, Section 5), the supervisor **drops all demands and raises a critical alarm but keeps petting**. Resetting the chip would not revive the bus — and the controller’s own silence already drops the calls equipment-side (7.2); a watchdog starvation here would only reset-loop the processor.

Boot gate. Every boot starts with demands hard-blocked: the supervisor holds a **boot gate** closed until the indoor sensor validates, setpoints are present, and the configuration CRC checks out. While the gate is closed the sensor-invalid rule above is suspended (otherwise the chip could never survive its own sensor bring-up); once the gate opens — once, latched — a later sensor loss stops the petting per the table above. If the gate is still closed after the boot grace period (default 120 s), a boot-grace alarm is raised. This implements the 7.3 boot-validation rule.

The supervisor also owns the **reset-loop accounting** (7.3) and the alarm registry behind the health entity and last-error diagnostics in Section 9. Recovery from *any* supervisor-driven demand drop still passes through the compressor minimum-off timer — nothing in the supervisor bypasses the compressor protections.

7.5 The external hardware watchdog

An independent supervisor (TPL5010 / MAX6369 class) that the controller's safety task must pet periodically. The safety task pets it **only when its own invariants hold** — a wedged UI or control task cannot keep it fed.

On timeout the watchdog — in hardware, with no firmware involvement — **forces the no-demand state**:

1. **Holds the RS-485 driver-enable (DE) off** → the controller cannot drive the bus; its demands age out and equipment drops the calls.
2. **Cuts the common relay-coil feed (Case B)** → all output relays open regardless of GPIO states.

Both actions together are what "watchdog forces no-demand" means; the DE cut alone is not sufficient once relays exist. The watchdog **remains armed at all times**, including during firmware updates — there is no "disable for update" mode.

The SlyTherm's processor-internal watchdog only reboots the chip; it is a convenience layer, not the safety layer. The layered set is: internal task watchdog (hung task) → external hardware watchdog (dead chip) → equipment comms-loss timeout (catch-all).

7.6 UI failure is a comfort outage, not a safety event

The touchscreen UI runs isolated from the protocol/safety tasks. A UI crash leaves the safety task, the demand-refresh discipline, and the watchdog chain untouched; a whole-chip hang lands on the external watchdog as above. The worst credible UI-load failure is a delayed demand refresh — which degrades toward *no demand*, the safe direction.

Reduced safe-UI (crash-loop containment). A non-critical rendering fault must degrade, not boot-loop the panel. When the reset-loop lockout latches (7.3: ≥ 3 abnormal resets in 30 min — the control side is already at no-demand), the next boot builds a **minimal known-good screen** — plain temperature, mode, and alarms, with the trend graph, mode-tint gradient, and RS-485 LISTEN capture switched off — instead of re-running the UI that crashed. The reduced state is latched in NVS and **persists until manually cleared** by the on-screen **“Restore full screen”** button, which clears both the reduced-UI flag and the reset-loop history and reboots (a software reset, which does not itself re-latch). This is a UI-resilience layer only; control and safety already failed to no-demand on the same latch.

8. Configuration Parameter Reference

All control and protection parameters are **firmware-resident**: they live on the controller, survive network outages, and are exposed to Home Assistant as editable `number` entities (Section 9). Home Assistant edits them but never owns them. Values written outside the documented range are **clamped or rejected** by the firmware. The defaults below are the boot / factory-reset values.

Temperatures are °C; durations are seconds unless noted. The “Constant” column gives the firmware identifier for cross-reference with engineering documentation.

8.1 Compressor protection

These are the **primary** compressor protections — do not widen them on the assumption that the outdoor unit protects itself (Section 7.1).

Parameter	Constant	Default	Range / notes
Compressor minimum OFF time	`kCompressorMinOffS`	**300 s**	240–900 s. The outdoor unit's internal ~3-min delay is a backstop only
Compressor minimum ON time	`kCompressorMinOnS`	**300 s**	60–1200 s
Maximum compressor starts per hour	`kCompressorMaxStartsPerH`	**3**	—
Post-boot compressor hold-off	`kBootCompressorHoldoffS`	**300 s** (+ 0–60 s random jitter)	Timer state is persisted; if it is missing/invalid at boot, the full hold-off is enforced
Reset-loop lockout count	`kResetLoopLockoutCount`	**3 resets**	≥ 3 watchdog/brownout resets…
Reset-loop window	`kResetLoopWindowS`	**1800 s** (30 min)	…within this window → latched NO-DEMAND, manual clear required

8.2 Setpoints, deadband, changeover, presets, and holds

Parameter	Constant	Default	Range / notes
Call hysteresis (heat/cool differential)	`kCallHysteresisC`	**0.55 °C** (≈ 1 °F)	A 0.5 °F differential is field-verified to short-cycle; 0.3 °C enter / 0.2 °C exit is an acceptable alternative
Auto-mode heat↔cool setpoint deadband	`kMinSetpointDeltaC`	**2.8 °C** (5 °F)	Minimum separation between heat and cool setpoints
Deadband hard floor	`kMinSetpointDeltaFloorC`	**1.1 °C** (2 °F)	Firmware rejects/clamps violating writes and pushes the other setpoint
Auto-changeover dwell	`kChangeoverDwellS`	**1800 s** (30 min)	Minimum time since the opposite call; compressor min-off must also be satisfied
Changeover trigger sustain	`kChangeoverSustainS`	**600 s** (10 min)	The changeover trigger must persist this long
Manual-change hold type	`kDefaultHoldType`	**until next preset**	A manual setpoint/mode change creates a hold of this type; scheduled preset writes are ignored while a hold is active. Types: until-next-preset / 2 h / 4 h / indefinite
Timed hold, short	`kHoldShortS`	**7200 s** (2 h)	Timed-hold expiry does **not** revert setpoints — the

Parameter	Constant	Default	Range / notes
			next scheduled preset write restores the schedule
Timed hold, long	`kHoldLongS`	**14400 s** (4 h)	—
Preset roster size	`kMaxPresets`	**8 entries**	Roster is config-driven from Home Assistant (retained config topic); invalid entries are skipped
Preset name length	`kPresetNameMaxLen`	**23 characters**	A preset pair violating the deadband resolves with the cool value winning (heat pushed down)

8.3 Dual fuel (balance point, lockouts, escalation)

Parameter	Constant	Default	Range / notes
Balance point (economic changeover)	`kBalancePointC`	**−8 °C**	Mirrors the OEM (R02P034 “P124”) range −30…+15 °C
Balance-point hysteresis	`kBalancePointHystC`	**2 °C**	—
Compressor low-ambient lockout	`kCompressorMinOatC`	**−20 °C**	**Set from the installed model’s submittal sheet** (FLEXX-class rated to −30 °C); range disabled…−30 °C
Gas/aux high-ambient lockout	`kAuxMaxOatC`	**+10 °C**	No gas/aux heat above this outdoor temperature
Escalation droop	`kEscalationDroopC`	**1.0 °C**	Room droop below setpoint…
Escalation time	`kEscalationMinS`	**1800 s** (30 min)	…sustained this long…
Escalation HP-demand threshold	`kEscalationHpDemandPct`	**95 %**	…while heat-pump demand is saturated → stage gas
De-escalation time	`kDeescalationMinS`	**3600 s** (60 min)	Stage back to heat pump after this, with outdoor temp above balance point + hysteresis

Hard validation rule: the firmware rejects any lockout/balance-point combination that leaves **any outdoor-temperature band with no permitted heat source**. A configuration that can silently produce “no heat at −25 °C” is itself treated as a fault.

8.4 Gas modulation

Parameter	Constant	Default	Range / notes
Modulation floor (low fire)	`kGasFloorPct`	**40 %**	Chinook documented low fire. Valid demand is 0 or 40–100 % — never in between; sub-floor demand snaps to 0/low-fire with hysteresis
Gas maximum continuous runtime	`kGasMaxRuntimeS`	**14400 s** (4 h)	Without progress → drop + alarm. The heat pump has **no** runtime cap — alarm only, never auto-cycled
Gas minimum ON time	`kGasMinOnS`	**300 s**	60–900 s. Gates **comfort** stops only — the burner is held at low fire until the timer is served. Safety stops (sensor fault, invariant trip, maximum runtime, watchdog) extinguish the burner immediately
Gas minimum OFF time	`kGasMinOffS`	**300 s**	60–900 s. After a power cycle the OFF timer restarts fresh unless persisted state proves it was already served

8.5 Defrost tempering

Parameter	Constant	Default	Range / notes
Tempering heat demand	`kDefrostTemperHeatPct`	**35 %** fixed	Never PID-driven
Tempering time cap	`kDefrostTemperMaxS`	**900 s** (15 min)	Hard cap; fan % and heat % separately configurable

> ⚠ **Pending verification:** ownership of defrost tempering (whether the interface board handles it autonomously in Case A) is unconfirmed. The expected outcome is that the controller only detects defrost and holds steady; these parameters bound the alternative.

8.6 Heat-pump demand shaping

Parameter	Constant	Default	Range / notes
Demand slew rate	`kHpSlewPctPerMin`	**10 %/min**	Applies only if a proportional demand path exists
Demand step size	`kHpStepPct`	**5 %**	—
Heat-pump demand floor	`kHpFloorPct`	**30 %**	**Verify against the installed model**

> ⚠ **Pending verification:** whether the installed system offers a proportional (modulating) heat-pump demand path at all — vs staged demand — is unconfirmed. In the staged/relay case these parameters are unused; the relay path duty-cycles within the Section 8.1 timers instead.

8.7 Indoor sensor fusion

Parameter	Constant	Default	Range / notes
Per-sensor staleness timeout	`kSensorMaxAgeS`	**300 s**	180–900 s; relies on the 60 s bridge heartbeat (below)
Bridge heartbeat interval	`kSensorHeartbeatS`	**60 s**	Home Assistant republish cadence (Section 9.4)
Valid range, low	`kSensorRangeMinC`	**5 °C**	Readings outside the gate are rejected
Valid range, high	`kSensorRangeMaxC`	**40 °C**	—
Outlier exclusion	`kSensorOutlierC`	**4 °C**	A sensor > 4 °C from the participant median is excluded + alarmed
Occupied sensor weight	`kOccupiedWeight`	**2.0**	vs 1.0 unoccupied (“follow-me” weighting)
Occupancy window	`kOccupancyWindowS`	**1800 s** (30 min)	—
Weight ramp time constant	`kWeightRampTauMinS` / `MaxS`	**600–1800 s**	Sensors phase in/out smoothly
Fusion smoothing time constant	`kFusionSmoothingTauMinS` / `MaxS`	**120–300 s**	—
Fusion slew limit	`kFusionSlewCPerMin`	**0.1 °C/min**	A sensor joining/leaving cannot step the control input
Fallback-sensor disagreement alarm	`kDs18b20DisagreeAlarmC`	**5 °C**	Local sensor vs fused aggregate sanity check
Per-sensor calibration offset	`kSensorOffsetMaxC` (clamp)	**0 °C**	Adjustable **±5 °C** per sensor (0.1 °C steps in Home Assistant), including the built-in fallback

Parameter	Constant	Default	Range / notes
			sensor. Applied **before** the health gates, so range/stuck/outlier checks judge corrected values; an offset change ramps through the fusion slew limit rather than stepping the control input

8.8 Outdoor temperature

Parameter	Constant	Default	Range / notes
Source staleness	`kOatStaleS`	**1800 s** (30 min)	→ fall to the next source rung: bus sensor → wired outdoor sensor → Home Assistant weather
Cross-source disagreement alarm	`kOatRungDisagreeAlarmC`	**5 °C**	—

All outdoor sources stale → **fail cold**: gas heat allowed, compressor locked out (and cooling locked out under the indoor-temperature policy below). The compressor is never run on an unknown outdoor temperature.

8.9 Fallback and degraded modes

Parameter	Constant	Default	Range / notes
Network-stale fallback, heat-to	`kFallbackHeatSetpointC`	**18 °C**	Dual-bounded with the cool-to value; mode = last user mode; never escalates to OFF
Network-stale fallback, cool-to	`kFallbackCoolSetpointC`	**27 °C**	27–28 °C acceptable
Network staleness threshold	`kMqttStaleS`	**1800 s** (30 min)	No MQTT command/heartbeat traffic for this long → fallback profile
Degraded-mode heat floor	`kDegradedHeatFloorC` / `kDegradedHeatCeilC`	**16–18 °C**	Local-fallback-sensor-only mode (only when a local sensor is fitted — see below): bounded heat, **cooling disabled** (or ≥ 29 °C ceiling), demand capped, persistent alarm
Indoor cooling lockout	`kCoolingIndoorLockoutC`	**18 °C**	Never cool when indoor temperature is below this

> **No local sensor on this wall unit (default build).** The on-board DS18B20 / local fallback slot is compiled **off** (`SLYTHERM_LOCAL_SENSOR` default off; `-DSLYTHERM_DS18B20` not built). The room temperature comes **only** from the MQTT/HA room sensors. Consequences: the **degraded / local-fallback mode** above and the **fallback-sensor disagreement alarm** and **built-in-fallback offset** in §8.7 **do not apply** to this build; and **all room sensors stale → fail to no-demand** (0 on every channel + alarm) — there is no local heat-floor backstop. The wall unit's temperature availability depends entirely on the MQTT bridge + Wi-Fi (see the User Manual, *Remote sensors*, and docs/04 §4). The local paths remain **compiled-in behind the flag** for other hardware.

8.10 Wall-screen lock

Parameter	Constant	Default	Range / notes
Auto-relock after inactivity	`kUiAutoRelockS`	**120 s**	30–600 s. No touch for this long → the screen relocks (and any installer-settings session expires)
PIN attempts before backoff	`kUiPinMaxAttempts`	**5**	3–10 failed entries…
PIN entry backoff	`kUiPinBackoffS`	**60 s**	…then entry is refused for this long (30–600 s) before attempts reset

The lock gates **change intents only** — alarms, current temperature, and equipment status are never hidden at any lock level, and the lock is tamper resistance, not a safety layer (forgotten-PIN recovery: Section 9.8).

8.11 Smart recovery (pre-heat / pre-cool)

Disabled by default (`kRecoveryEnabledDefault` = false) until field-tuned; requires the Home Assistant `next_target` feed (Section 9.7). Advisory only — an early start is a normal call and passes every protection in 8.1.

Parameter	Constant	Default	Range / notes
Ramp-rate seed, heating	`kRecoverySeedHeatCPerH`	**1.0 °C/h**	Starting estimate per {mode, equipment} channel, used until that channel has learned from real run segments
Ramp-rate seed, cooling	`kRecoverySeedCoolCPerH`	**0.8 °C/h**	—
Maximum early-start lookahead	`kRecoveryMaxLookaheadS`	**7200 s** (2 h)	Hard cap on any early-start recommendation, whatever the learned rate suggests

8.12 Relay output sequencing (Case B)

Parameter	Constant	Default	Range / notes
Minimum relay transition spacing	`kRelayMinTransitionMs`	**500 ms**	Minimum spacing between successive output transitions (Section 6.1). Never delays the failsafe all-off

8.13 Installer notes

- After any configuration change to lockouts or balance point, confirm the controller accepted it (the echoed state in Home Assistant shows the post-clamp value) and that no validation alarm is raised.
- Set `kCompressorMinOatC` from the **installed outdoor unit's submittal sheet**, recorded during the pre-install survey — not from the default.
- Record the as-left value of every parameter you change in the commissioning record (Section 10).

9. Home Assistant Integration

The SlyTherm integrates with Home Assistant (HA) over **MQTT on the local network only** — no cloud service is used or required. HA is a **supervisory** layer: dashboards, schedules, remote sensors, history, and phone access. Every control HA offers also works (or degrades safely) when HA is down; **loss of HA or the broker never raises any demand**.

9.1 Prerequisites

Requirement	Notes
MQTT broker	A local broker reachable from the controller and HA (e.g., the Mosquitto broker add-on commonly used with Home Assistant)
HA MQTT integration	Enabled, with **MQTT Discovery** active (default discovery prefix `homeassistant`)
Wi-Fi	The controller joins the local Wi-Fi; credentials are provisioned without pulling the unit from the wall (captive portal / Bluetooth provisioning)
Remote phone access (optional)	Self-hosted WireGuard or Tailscale with the HA Companion app; no cloud relay required

9.2 Discovery and entities

The controller publishes a **device-based MQTT Discovery payload** on boot; HA auto-creates one device with the following entities:

Entity	Type	Purpose
`climate.slytherm_hvac`	climate	Modes off/heat/cool/heat_cool, single + dual setpoints, current temperature, action
`sensor.slytherm_active_equipment`	sensor	`idle` / `hp_heat` / `gas_heat` / `cool` / `defrost` — distinguishes heat-pump heat from gas heat (HA's `hvac_action` cannot)
`sensor.slytherm_modulation`	sensor (%)	Live gas modulation: 0 or **40–100 %** (there is no 1–39 % band)
`sensor.slytherm_outdoor_temp`	sensor (°C)	Outdoor temperature used by the dual-fuel logic
`sensor.slytherm_outdoor_source`	sensor	Which source supplied it: `bus` / `wired` / `ha` / `none`
`sensor.slytherm_fusion`	sensor (JSON attributes)	Effective room temperature, participating sensors, occupancy weighting
`sensor.slytherm_compressor_min_off_remaining`	sensor (s)	Compressor minimum-off countdown
`binary_sensor.slytherm_compressor_locked_out`	binary_sensor	Compressor lockout active (timers, low outdoor temp, fault)
`sensor.slytherm_changeover_reason`	sensor	Why the last heat↔cool / gas↔HP changeover happened
`sensor.slytherm_blower`	sensor	Blower/inducer state
`sensor.slytherm_fault`	sensor	Decoded equipment fault code/text

Entity	Type	Purpose
`binary_sensor.slytherm_health`	binary_sensor (problem)	Controller health (Wi-Fi/MQTT/sensor/watchdog)
`sensor.slytherm_last_error`	sensor (diagnostic)	Last error string
`sensor.slytherm_hold`	sensor (diagnostic)	Active hold type (`none` / `until_next_preset` / `two_hours` / `four_hours` / `indefinite`); remaining seconds as an attribute
`sensor.slytherm_lock`	sensor (diagnostic)	Wall-screen lock state (`unlocked` / `user_locked` / `installer_locked`); lock level and whether a PIN is set as attributes (Section 9.8)
`switch.slytherm_em_heat`	switch	EM HEAT (gas-only emergency heat): `ON` engages, `OFF` restores the prior mode. Deliberately a switch — not an HVAC mode and not a preset, so a scheduled preset write can never silently disengage it
Per-remote-sensor diagnostics	sensor / binary_sensor	Per-sensor age and fusion participation (diagnostic category)
`number.slytherm_sensor_<id>_offset`	number (config)	Per-sensor calibration offset, ±5 °C in 0.1 °C steps — includes the built-in fallback sensor (id `local`)
Configuration numbers	number	Every parameter in Section 8, range-clamped by firmware

9.3 Topic map (integrator reference)

Commands (HA → controller), all under `slytherm/cmd/`:

Topic	Payload
`slytherm/cmd/setpoint`	float °C (heat or cool mode)
`slytherm/cmd/target_temp_low` / `target_temp_high`	float °C (heat_cool mode dual setpoints)
`slytherm/cmd/mode`	`off` / `heat` / `cool` / `heat_cool`
`slytherm/cmd/fan_mode`	`auto` / `on` / `circulate`
`slytherm/cmd/preset`	a preset name from the configured roster (default `home` / `away` / `sleep`)
`slytherm/cmd/hold`	`until_next_preset` / `two_hours` / `four_hours` / `indefinite` / `clear`
`slytherm/cmd/em_heat`	`ON` / `OFF`
`slytherm/cmd/sensor/<id>/offset`	float °C within ±5 (per-sensor calibration; `local` = built-in fallback sensor)
`slytherm/cmd/outdoor_temp`	float °C — HA-weather bridge feeding the outdoor-temperature ladder's third rung (`ha` source); plausibility-gated −50…55 °C; republish at least every 30 min or the rung goes stale (Section 8.8 fail-cold policy applies)
`slytherm/cmd/next_target`	JSON `{"temp": 21.0, "mode": "heat", "in_s": 5400}` — the next scheduled setpoint change, for smart recovery (Section 9.7); all three keys required, invalid payloads ignored
`slytherm/cmd/lock_clear`	exactly `clear_user_pin` — forgotten-PIN recovery (Section 9.8); any other payload, including empty, is ignored

State (controller → HA), all under `slytherm/state/` and echoing the **post-validation** values: `current_temp`, `setpoint`, `target_temp_low`, `target_temp_high`, `mode`, `fan_mode`, `preset`, `hold` (JSON: type + remaining seconds), `em_heat`, `action` (`off` / `idle` / `heating` / `cooling` / `fan` / `defrosting`), `active_equipment`, `modulation`, `outdoor_temp`, `outdoor_source`, `fusion`, `compressor_min_off_remaining`, `compressor_locked_out`, `sensor/<id>/offset`, `changeover_reason`, `fault`, `blower`, `health`, `last_error`, `lock` (JSON: state, level, whether a PIN is set).

Important contract points:

- **Setpoint clamping is firmware-side.** The dual-setpoint deadband (cool ≥ heat + 2.8 °C default, 1.1 °C hard floor) is enforced on the controller: violating writes are clamped and the other setpoint is pushed; HA sees the corrected values echoed on the state topics.
- **Availability/LWT:** the controller publishes `slytherm/availability` = `online`; its MQTT **Last Will** sets `offline` so HA marks every entity unavailable the moment the controller drops off the broker.
- A rejected or malformed command payload never mutates controller state.

9.4 Remote room sensors (temperature + occupancy)

Remote sensors (Zigbee/ESPHome devices known to HA) reach the controller via an **HA bridge automation** that republishes them to MQTT:

- **Per-sensor state topic:** `slytherm/sensors/<id>/state`, JSON payload `{"temp": <°C>, "occ": true|false|null, "bat": 0-100|null, "hum": <%>|null}`.
- **Publish non-retained**, on change **and on a 60-second heartbeat**. The firmware's per-sensor staleness timeout (default 300 s) requires the live cadence; non-retained publication ensures a dead broker cannot replay stale temperatures.
- **Per-sensor presence topic:** retained publication at `slytherm/sensors/<id>/presence`, carrying the sensor's last-seen or present/away state. Because it is retained, it **seeds the controller's sticky home/away on boot** — a rebooting controller recovers each sensor's presence immediately, before the first live heartbeat arrives.
- **Roster/config topic:** retained JSON at `slytherm/config/sensors`. Each sensor entry carries its **`id`**, a friendly **`name`** (the display name shown on the wall Sensors screen), which presets the sensor participates in, and per-sensor staleness overrides.

Sensor **fusion happens in the controller, not in HA templates** — the control input must survive an HA outage, and per-sensor health must be visible to the controller's safety layer. Do not build an averaging template in HA and feed it as a single sensor.

Sensor hardware note: use Zigbee temperature sensors (e.g., SONOFF SNZB-02D, Aqara) and mmWave occupancy sensors (e.g., Aqara FP2, Apollo MSR-2) integrated through HA. **Ecobee remote sensors cannot be used** — their 915 MHz protocol is proprietary and unreceivable by this hardware.

9.5 Schedules and presets

- **HA owns schedules.** The HA scheduler or automations drive the thermostat by writing presets and/or setpoints. The controller does not store a weekly schedule.
- **The preset roster is configuration, not firmware.** Publish retained JSON at `slytherm/config/presets` — `{"presets":[{"name":"home","heat":21.0,"cool":25.0},…]}` — up to **8 entries**, names ≤ 23 characters, setpoints within the climate limits; invalid entries are skipped, and the climate entity's preset list is rebuilt from the roster (default `home` / `away` / `sleep`).
- **Holds:** a manual setpoint/mode change creates a hold of the configured default type (factory: until next preset). While a hold is active, scheduled preset writes are ignored; timed holds expire by clock and the next scheduled preset write restores the schedule (expiry never reverts setpoints on its own). See Section 8.2 for the timer values.
- **Starter package:** a ready-made HA package (weekly schedule, vacation calendar, runtime-based filter reminder, temperature/humidity alerts, presence-based Away) ships in the project's `ha/` directory with its own install instructions (`ha/README.md`). Load it at commissioning rather than hand-building these automations.
- **The controller owns the outage fallback.** If MQTT goes stale (> 30 min): heat-to 18 °C / cool-to 27 °C, mode = last user mode — never escalated to OFF, and never a single bare setpoint.

9.6 Safety interactions (binding on the integrator)

- HA is for **comfort and visibility, never a safety layer**. All failsafes live on the controller, the furnace IFC, and the equipment protections. Do not build HA automations that assume they are load-bearing for safety.
- Loss of HA/MQTT never raises demand of any kind; recovery from MQTT/sensor faults honors the compressor minimum-off timer.
- Health and diagnostics are published so silent failures are visible in HA (and on the touchscreen and status indication).
- As a diagnostic fallback, the controller serves a local web page usable when HA or the broker is down. It is a fallback, not a primary UI.

9.7 Smart recovery — the `next_target` contract

Smart recovery (pre-heat/pre-cool, Section 8.11) needs to know the *next* scheduled setpoint before it arrives. Because HA owns the schedule (Section 9.5), HA supplies that look-ahead:

- **Publish the next scheduled target** to `slytherm/cmd/next_target` as JSON `{"temp": 21.0, "mode": "heat", "in_s": 5400}` — the upcoming setpoint, which side it serves (`heat` / `cool`), and the seconds until it takes effect. Republish whenever the schedule or the remaining time changes (piggyback on the sensor-bridge heartbeat, Section 9.4).
- **The parse is tolerant:** all three keys are required (`in_s` capped at 7 days); an invalid payload is ignored, and publishing nothing simply means no pre-start. HA stays optional, as always.
- **Advisory only:** the recovery estimator recommends an early start; the control logic decides, and the compressor timers, dual-fuel arbitration, and every other protection still gate the resulting demand. A bogus `next_target` can at worst start a normal, fully protected call early — it can never bypass a lockout or timer.

9.8 Screen lock and forgotten-PIN recovery

The wall screen's PIN lock (user-facing description in the User Manual; parameters in Section 8.10) surfaces in HA two ways:

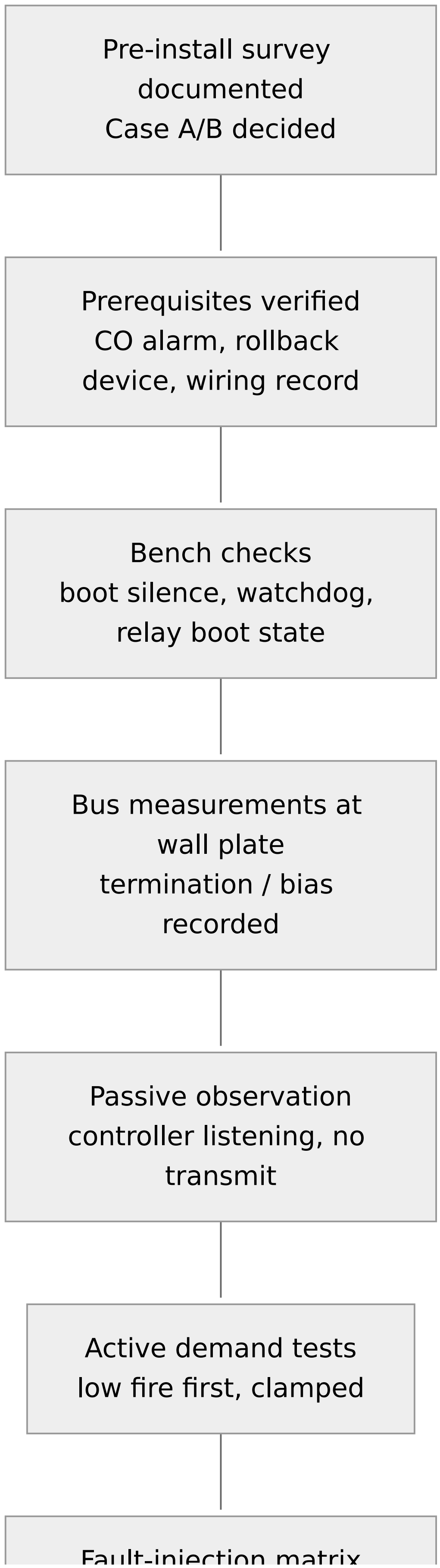
- **Visibility:** `slytherm/state/lock` → `sensor.slytherm_lock` (diagnostic) carries the lock state, lock level, and whether a PIN is set, so a locked-out (or installer-locked) thermostat is diagnosable remotely.
- **Forgotten-PIN recovery:** publish exactly `clear_user_pin` to `slytherm/cmd/lock_clear`. The user PIN is cleared and a user lock released — **no PIN required**. Rationale: anyone who can publish to the broker already has full climate control (every setpoint, mode, and EM HEAT), so **HA/broker access is admin by definition** — a PIN gate on this topic would add no security, only a second thing to forget. The lock is tamper resistance, never a safety layer.
- **The installer code is *not* clearable over MQTT.** Recovery from a lost installer code is physical (USB reflash / NVS wipe at the wall). An NVS wipe fails *open* — lock disabled, no PINs — so a wipe can never brick the wall UI.

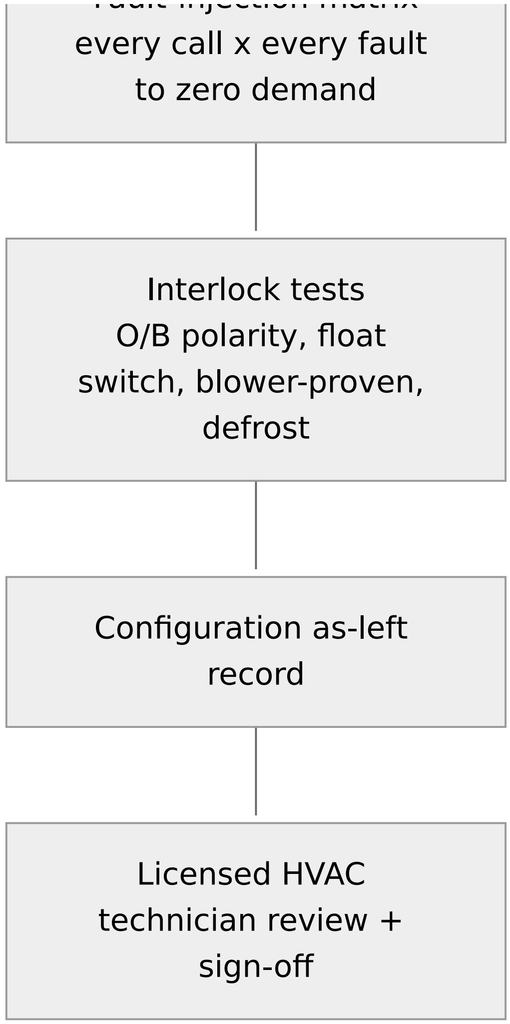
10. Commissioning

Commissioning is **not optional**. The SlyTherm’s safety case rests on behaviors that must be **proven on the installed equipment** — most importantly that every demand channel reaches zero under every induced fault. The system must not be left in service until every item below is signed off, and a licensed HVAC technician has reviewed the installation.

Bring: oscilloscope or logic analyzer (for the boot-silence and relay boot-state checks), multimeter, and a means of inducing the faults below. Record every result; the completed tables in 10.6 are the commissioning record.

10.1 Commissioning sequence


Pre-install survey documented
Case A/B decided
Prerequisites verified
CO alarm, rollback device, wiring record
Bench checks
boot silence, watchdog, relay boot state
Bus measurements at wall plate
termination / bias recorded
Passive observation
controller listening, no transmit
Active demand tests
low fire first, clamped
Fault-injection matrix



10.2 Prerequisites (verify before any active test)

#	Prerequisite	✔
P1	Pre-install survey complete and documented: architecture (Case A/B), thermostat and outdoor-unit models, IFC DIP states, full low-voltage wiring map, wall-plate conductor count	☐
P2	CO alarm installed in the dwelling and tested	☐
P3	OEM thermostat retained on site; rollback procedure (Section 11) known **and rehearsed**	☐

#	Prerequisite	✔
P4	Bus termination/bias measurements at the wall plate recorded (Section 5.5)	☐
P5	Demand-message format confirmed from captures of the installed equipment (engineering gate — see Pending-verification notes); controller verified silent-listening until then	☐
P6	Confirmed by observation/capture that the furnace IFC enforces flame/limit/pressure independently of the bus	☐
P7	OEM communicating thermostat **removed from the bus** before any controller transmission (one master only)	☐

10.3 Bench / boot-state checks

> **Note — bench mode:** the thermostat firmware's **default build is demands-disabled**: it runs the full control loop over MQTT only, with the actuator stubbed to a logger — demands are printed, never transmitted, no bus TX and no relay GPIO is enabled. This lets you exercise sensors, Home Assistant discovery, and the whole control pipeline on the bench with nothing connected. Live demand output requires the deliberate build-time gates described in the firmware source (`src/main_thermostat.cpp`).

C1 — Boot silence (scope check). 1. Connect the scope across the bus A–B at the controller tap. 2. Power-cycle and hard-reset the controller several times. 3. **Pass:** not one byte appears on the bus during boot/reset; the DE line stays in the receive/idle state (resistor pull) throughout.

C2 — Hardware watchdog forces no-demand (hang test). 1. Use the firmware's test hook (or halt the processor) to stop watchdog petting while a heat call is active. 2. **Pass:** on timeout the watchdog hardware forces DE off (bus goes silent) **and** (Case B) cuts the relay-coil feed (all relays open) — and the equipment subsequently drops the call.

C3 — Relay boot state (Case B). 1. Scope (or watch the sense inputs on) all relay outputs through several power-ups, resets, and brownouts. 2. **Pass:** every relay coil remains de-energized through boot/reset; no contact closes transiently.

C4 — Reset-loop lockout. 1. Force 3 resets within the 30-minute window. 2. **Pass:** the controller latches NO-DEMAND and requires a manual clear.

C5 — Brownout behavior. 1. Sag the supply mid-call; restore. 2. **Pass:** controller boots to no-demand, validates mode/setpoints/sensors before any demand, cross-checks the restored mode against outdoor lockouts, and enforces the compressor hold-off if timer state was lost.

C6 — Compressor minimum-off across reboot. 1. Stop a compressor call, then power-cycle the controller during the minimum-off period. 2. **Pass:** after reboot the hold-off resumes (or restarts in full); no compressor demand is emitted early.

C7 — Mutual-exclusion invariant. 1. Using the firmware test hook, attempt to command gas heat and compressor heat simultaneously. 2. **Pass:** the controller zeros all channels and raises an alarm.

C8 — Firmware-update policy. 1. Attempt an over-the-air update while a demand is active → **must be refused/deferred**. 2. Confirm the watchdog stays armed throughout an accepted update, and exercise the automatic rollback (unconfirmed image → previous firmware) once on the bench.

10.4 The fault-injection matrix (core acceptance test)

For **each equipment call** {furnace heat, HP heat, HP cool, aux/backup heat}, induce **each fault** below mid-call and confirm: the call drops to **zero demand** (bus silent / relays open, equipment ramps down via its own logic), and recovery honors the compressor timers.

↓ fault call →	Furnace heat	HP heat	HP cool	Aux/backup
Pull bus mid-call	☐	☐	☐	☐
Controller hang (HW watchdog)	☐	☐	☐	☐
Sensor fault / flap	☐	☐	☐	☐
MQTT loss	☐	☐	☐	☐
Brownout / power-cycle mid-call	☐	☐	☐	☐

Per-row procedures:

- **Pull bus mid-call:** with the call active and stable, disconnect the controller's bus tap (terminals 1/2). Observe at the equipment that the call drops within its communications-loss timeout. **This row also answers a standing open question:** confirm silence drops the call at **both** the furnace and the heat-pump/interface-board side — the bus's coordinator topology may not propagate silence to the non-coordinator unit. ⚠ **Any cell that fails means that channel needs a hardware call-removal path before the system may be left in service** (Section 1.7).
- **Controller hang:** as test C2, repeated per active call.
- **Sensor fault / flap:** unplug the local fallback sensor, kill the remote sensors (stop the HA bridge), and flap a remote sensor in and out. On quorum loss the active call must drop to zero with an alarm; a flapping sensor must **not** short-cycle the compressor through the failsafe path (recovery waits out minimum-off).
- **MQTT loss:** stop the broker mid-call. No demand may *rise*; after the staleness window the fallback setpoint profile governs; the active call behaves per that profile. Restore and confirm clean recovery.
- **Brownout / power-cycle mid-call:** as test C5, repeated per active call.

10.5 Equipment interlock tests

E1 — O/B reversing-valve polarity. 1. With the compressor **idle**, set the configured polarity (default: B = energized in heating — Gree convention). 2. Command a heat-pump heat call; verify warm supply air / correct refrigerant direction. Command cooling; verify cooling. 3. **Pass:** heating heats and cooling cools. A wrong guess inverts both. Never switch O/B with the compressor running.

E2 — Condensate float switch (cooling enabled, either case). 1. Lift/trip the float manually during an active cooling call. 2. **Pass:** the cooling call is broken **with the firmware unaware** — the hardwired series path opens the call regardless of software; any sense input is for alarming only.

E3 — Blower-proven interlock. 1. Command cooling while preventing/blocking blower confirmation (e.g., suppress the bus blower telemetry or the G/Y feedback per case). 2. **Pass:** the cooling call is refused or dropped without blower confirmation.

E4 — Defrost ride-through. 1. Force a defrost cycle from the outdoor unit's service procedure ("FO 3 Force Cycle" class). 2. **Pass:** the controller detects defrost (bus signature or D/W sense), holds steady, does not disrupt the cycle, and any tempering stays within the configured fixed demand and 15-minute cap. Record who commanded tempering (controller vs interface board) — this closes a documented open question.

E5 — Demand-discipline spot checks. Confirm the demand clamp (gas 0 or 40–100 % only), per-equipment maximum runtime policy, anti-short-cycle timers, and per-channel demand refresh are active (firmware diagnostics page / HA diagnostics).

10.6 Final sign-off

#	Item	Result / value	Initials
S1	Prerequisites P1–P7 complete		
S2	C1 boot silence — scope-verified		
S3	C2 watchdog no-demand (DE off + coil-cut if Case B)		
S4	C3 relay boot state (Case B)		
S5	C4 reset-loop lockout		
S6	C5 brownout-to-safe		
S7	C6 min-off across reboot		
S8	C7 mutual exclusion		
S9	C8 update policy + rollback exercised		
S10	10.4 matrix — **all 20 cells** reach zero demand		
S11	E1 O/B polarity verified on installed equipment		
S12	E2 float switch kills cooling, firmware unaware		
S13	E3 blower-proven interlock		
S14	E4 defrost observed and not disrupted		
S15	E5 demand-discipline checks		

#	Item	Result / value	Initials
S16	Configuration as-left values recorded (Section 8)		
S17	Bus termination/bias measurements recorded (Section 5.5)		
S18	Homeowner briefed: alarms, loss-of-heat behavior, rollback		
S19	**Licensed HVAC technician review** — name/licence recorded		

11. Rollback to the OEM Thermostat

Rollback returns the system to its **certified configuration**. The wiring must be designed (Section 3.2: labeled conductors, ferruled terminations, accessible terminal blocks) so this swap **takes minutes**. Rehearse it once at commissioning.

11.1 When to roll back

- Sustained loss-of-heat alarm in heating season that cannot be promptly diagnosed.
- Any commissioning or in-service finding that a demand channel does not fail safe.
- Equipment service visits — present the technician with the certified configuration.
- Owner request, sale of the home, or decommissioning.

11.2 The rollback device

Installation	Rollback thermostat
Communicating (Case A)	**Dettson R02P034** (current successor to the discontinued R02P030; R02P032 also applies)
Conventional (Case B)	**Dettson R02P033** or other conventional thermostat per the original wiring map

The rollback device is kept **on site** as a condition of installation (Section 1.4).

11.3 Procedure

1. At the touchscreen or Home Assistant, set the system to **OFF**. Wait for `active_equipment` to read `idle`. If the controller is dead, skip this step — equipment will drop its calls on bus silence / open relays.
2. Kill 24 VAC to the control circuit (furnace service switch).
3. Remove the SlyTherm display from its bracket and disconnect the field wiring at the enclosure terminal blocks: R, C, 1, 2 — and, in Case B, Y1/(Y2)/ O-B/G and the sense conductors.
4. **Leave the hardwired condensate float switch in the Y circuit** — it is equipment protection, not part of the controller.

5. Remove any termination/bias components **you** added at the wall plate (per the Section 5.5 record) so the bus returns to its pre-install state.
6. Mount the OEM thermostat base and land the conductors per the **photographed pre-install wiring map** (survey step 5): R, C, 1, 2 for a communicating thermostat; the conventional map for an R02P033.
7. Restore 24 VAC. Verify the OEM thermostat powers up and communicates (communicating models) or controls conventionally.
8. Run one heat call (and one cool call, in season) to completion of startup to confirm normal operation.
9. In Home Assistant, the SlyTherm entities will go `offline` (Last Will); disable any automations that command them.

11.4 Returning to service

Re-installation of the SlyTherm after a rollback repeats the relevant parts of commissioning (Section 10): boot-silence check, the fault-injection rows for any channel whose wiring was touched, and the interlock tests if Case-B wiring was disturbed.

12. Specifications and Compliance

12.1 Electrical

Item	Specification
Supply	24 VAC nominal from the furnace control transformer (R/C); front end rated for ≥ 50 V peak (transformers run 28–30 VAC light-load)
Power consumption	≤ ~3 VA from the transformer (display board 1.6–2.25 W plus conversion losses)
Power conversion	Isolated AC-input AC/DC module, 9–36 VAC input, 5 V ≥ 1 A output; DC ground isolated from furnace C
Input protection	Inline slow-blow fuse 250–500 mA on R; MOV (~40–47 VAC working) across R–C; 470–1000 µF / 63 V, 105 °C bulk capacitance
Bus interface	EIA/TIA-485 2-wire half-duplex (CT-485), 9600 bps 8N1; 3.3 V transceiver with explicit driver-enable; TVS-protected (SM712/SMBJ class) with ~10 Ω series elements; stub < 0.3 m
Relay outputs (Case B)	Y1, Y2, O/B, G — 24 VAC-rated dry contacts, normally open, de-energized = no demand; common coil feed interrupted by the hardware watchdog
Sense inputs (Case B / instrumentation)	Opto-isolated 24 VAC sense: D/W, Y, G, W
Local sensors	DS18B20 1-Wire (indoor fallback, outdoor, optional supply-air); 4.7 kΩ pull-up; optional I2C humidity/temperature module
Supervision	External hardware watchdog (TPL5010/MAX6369 class): on timeout forces driver-enable off and (Case B) cuts relay-coil power

Item	Specification
Network	2.4 GHz Wi-Fi; MQTT over the local network; no cloud dependency

12.2 Environment

Item	Specification
Controller mounting	Indoor living space, at the thermostat wall location
Furnace-side enclosures	Mounted on the cool side / outside the furnace cabinet; polycarbonate or ≥ 105 °C-rated material near heat (ABS is ~80 °C continuous)
Operating temperature range	**TBD** — to be characterized before release; display-board vendor ratings govern in the interim
Outdoor sensor	Outdoor-rated cable; north wall, shaded

12.3 Compliance — honest statement

The SlyTherm is not a certified or listed product.

- It carries **no CSA, UL, or other safety listing**.
- The furnace and heat pump are certified **as systems with approved controls**; installing the SlyTherm in place of the OEM thermostat is a **modification of a certified gas appliance's control system** and places the installation **outside the certified configuration** — even though the design does not alter the combustion-safety chain (the certified IFC retains all combustion safeties, which are not reachable over the bus) or the refrigerant-circuit protections.
- This **can void the furnace and heat-pump warranties**, may **violate local gas or electrical code** (gas-appliance control work is often restricted to licensed technicians), and **could affect home insurance** after any fire, carbon-monoxide, or water incident traced to the controller.
- The product is **experimental and operated at the owner's risk** on a life-safety appliance. The supported alternative is a Dettson R02P034 thermostat with integration at the Home Assistant level.

Conditions of installation (CO alarm, retained OEM thermostat, licensed technician review, no safety-wiring modifications, documentation) are binding — see Section 1.4.

12.4 Design provenance and open items

This draft reflects a reviewed engineering design whose equipment-facing behaviors are **partially unverified**. The following remain open and are marked throughout the manual:

Open item	Where it matters
Installed architecture, Case A vs Case B (pre-install survey)	Sections 2, 6
Demand-message payload offsets (confirmed only from captures)	Section 5.7; transmit path is gated on it
Bus token/turnaround timing budget; single-unit qualification	Section 2.2
Mixed-mode IFC behavior (hardwired Y/G/W while communicating)	Section 6 (gates Case B)
Proportional vs staged heat-pump demand path	Section 8.6
Silence propagation to the non-coordinator unit, per channel	Sections 7.2, 10.4
Defrost tempering ownership	Sections 8.5, 10.5
In-wall code treatment of the AC/DC module	Section 3.2

Appendix — Control-Strategy Research (Cold-Climate Dual Fuel)

This appendix records the technical literature reviewed while designing the controller's dual-fuel heating strategy — both the approaches the product adopts and the approaches that were considered and deliberately **not** used, with the reasoning. It is reference material for installers and technically minded owners; nothing in this appendix changes an installation step. The working bibliography with implementation mapping is maintained in the repository (`docs/13-dual-fuel-control-research.md`).

Strategies the controller adopts (implemented or planned)

Economic switchover between heat pump and gas

Field and simulation studies of hybrid (heat pump + gas furnace) systems in cold climates consistently find that a **fixed** heat-pump/furnace switchover temperature is sub-optimal; the cost- and emissions-optimal changeover moves with the electricity-to-gas price ratio and the heat pump's efficiency curve versus outdoor temperature (ORNL/Purdue, Hybrid Heat Pump Controls; Smart Dual Fuel Switching System study, Ontario). Natural Resources Canada's instrumented hybrid-heating homes measured roughly a 30% seasonal greenhouse-gas reduction against furnace-only operation (NRCan). The controller therefore computes an *economic balance point* — run the heat pump whenever its efficiency at the current outdoor temperature beats the break-even point implied by configured energy prices — bounded by hard thermal floors (compressor lockout, capacity limits).

Predictive recovery ("start early, start gentle")

Rather than waiting for the room temperature to cross the setpoint and then demanding full heat, the controller predicts the crossing and begins a shallow, heat-pump-first recovery ramp early, holding gas in reserve behind a steeper fallback line — the adaptive-recovery scheme long established for heat-pump thermostats (US 5,622,310). Predictive control of this family shows double-digit heating-cost reductions with negligible comfort penalty in residential studies (Neurothermostat; Québec experimental-house predictive heating study).

Blower-first starts and low-speed pre-circulation

Ahead of a predicted heat call the controller may start the furnace's ECM blower on low for one to three minutes. Low-speed circulation destratifies the space (ceiling-warm/floor-cold mixing), which both improves comfort and gives the controller's fused room-temperature reading a truer picture *before* it commits to a heating stage (variable-speed airflow analyses). ECM power draw falls steeply at low speed, so the pre-run cost is single- digit watts. A short post-burn blower run to harvest residual heat from the exchanger is likewise standard practice.

Long, low-fire modulation

Modulating condensing furnaces earn their efficiency at part load: long, low-fire burns keep the heat exchanger condensing and avoid cycling losses and temperature overshoot (modulating-furnace fuel-utilization patent US 6,321,744). The controller's demand shaping is biased toward continuous low modulation rather than bang-bang cycling around the deadband.

Gas tempering during heat-pump defrost

During a defrost cycle the indoor coil runs cold; unmitigated, this produces the "cold blow" complaint typical of heat pumps, and defrost overhead adds 5–15% to energy use in humid subfreezing weather (defrost/supplemental-heat control, US 5,332,028). With a modulating furnace available, low-fire gas during defrost windows is the default tempering policy.

Seasonal self-calibration

Installed heat-pump performance deviates from nameplate in the field (ACEEE cold-climate monitoring; NRCan ccASHP assessment). The controller already records outdoor temperature, stage runtime, and room temperature trends; these are used to correct the assumed efficiency curve over a season, keeping the economic switchover honest without added instrumentation.

Strategies considered and NOT adopted

Fixed switchover temperature as the primary policy

The industry-default single changeover setpoint (commonly around −10 °C for a two-stage furnace in Canadian design conditions, eSim ASHP sizing study) was **rejected as the primary policy** for the reasons in the first section — it is retained only as an installer-configurable safety floor and manual override, never the optimizer.

Electric resistance auxiliary heat

Most published heat-pump recovery/defrost strategies assume electric strip auxiliaries. **Not applicable here**: this system's supplemental heat is the modulating gas furnace, which is both the cheaper marginal heat source and already in the airstream. Resistance strips would also add substantial electrical panel load that this installation class cannot spare.

Full model-predictive control (MPC) on the controller

Formal MPC — grey-box or neural building models optimized over a forecast horizon (intermittent-heating MPC) — shows the largest savings in the literature but was **deliberately not adopted on-device**: it requires reliable weather forecasts, a maintained building model, and optimization compute that do not belong on a safety-adjacent embedded controller. The adopted adaptive dual-ramp recovery captures the bulk of the benefit with a model the controller can learn and explain. MPC remains a candidate as a *supervisory* layer in Home Assistant, issuing ordinary setpoint schedules that the controller executes and re-validates.

Continuous 24/7 low-speed circulation

Running the blower continuously at low speed maximizes destratification but was **rejected as a default**: even efficient ECM blowers accumulate meaningful energy over a season, and constant airflow increases filter loading and perceived draft. The adopted policy is a scheduled circulation duty plus the predictive pre-runs described above, which deliver mixing when it matters.

Time-of-use / grid-responsive pre-conditioning

Pre-heating with the heat pump ahead of electricity price peaks (letting gas carry the peak) shows real savings in co-simulation (grid-responsive dual-fuel control; Canadian context in Heat Pumps Pay Off), and the mirrored summer strategy — model-predictive **pre-cooling** ahead of the peak window — has been demonstrated with ordinary smart thermostats (MPC optimal precooling).

Where the installation pairs time-of-use rates with **local solar and battery storage** (for example an EcoFlow Delta Pro Ultra), the economics strengthen further: the cheap-energy window is the utility off-peak band *plus* stored solar. In summer, on a plan whose peak begins at 07:00, it can pay to pre-cool the space **before 07:00 even while occupants are asleep** — compressor runtime happens on overnight rates or battery, and the home coasts through the morning peak. The same logic runs in reverse in winter: during extreme low-cost windows — ultra-low overnight rates, negative/curtailment pricing, or a battery holding surplus solar — the controller may pre-warm the house **above** the scheduled setpoint, banking heat in the building’s thermal mass while the heat pump is at its cheapest, so the expensive hours that follow are coasted through with neither compressor nor burner running. The controller treats both cases as scheduled economic actions bounded by configured comfort limits (they deliberately override the night/sleep comfort band within those limits), and consumes a simple “cheap-energy window” signal from Home Assistant rather than modeling the battery itself.

Deferred, not rejected: it requires the price/window feeds the controller does not yet consume. The recovery machinery is designed so price-shifted targets can be added later without new control paths.

Equipment resizing / oversizing guidance

Several studies optimize by re-sizing the heat pump itself. **Out of scope** for the controller — sizing is a system-design decision made before installation. Installers should consult the eSim Canadian sizing study during equipment selection.

13. Revision History

Rev	Date	Description
0.1	2026-06-11	Initial draft. PRE-RELEASE — NOT FOR FIELD USE. All “Pending verification” and “TBD” callouts open.
0.2	2026-06-11	Custom presets, timed holds, sensor calibration, gas cycle timers, HA starter package. PRE-RELEASE — NOT FOR FIELD USE; “Pending verification” and “TBD” callouts remain open.
0.3	2026-06-11	Screen lock, smart recovery, accessory coordination, safety supervisor, bench firmware. PRE-RELEASE — NOT FOR FIELD USE; “Pending verification” and “TBD” callouts remain open.
0.4	2026-07-06	Wall-UI refresh: MQTT/HA-only room sensors (no on-board DS18B20 — §8.9), hold-duration chooser + Home hold pill, System-tab 12 h trend graph, consistent green Wi-Fi/Home-system status + 12/24 h clock on Settings, auto-clearing recoverable alerts, and reduced safe-UI crash-loop containment (§7.6). PRE-RELEASE — NOT FOR FIELD USE; “Pending verification” and “TBD” callouts remain open.